

Estimating a self-excitation kernel from a series of events.

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1 Introduction

In *self-exciting* processes, past events occurrences facilitate the occurrence of future events. Many phenomena in real-life situations exhibit this self-exciting behavior; a big earthquake will produce changes in the tension of the ground making it more likely for aftershocks to occur [20], a popular tweet in social media about a trending topic will likely be redistributed [11], or an infectious disease might widely spread as more and more individuals become infected [15].

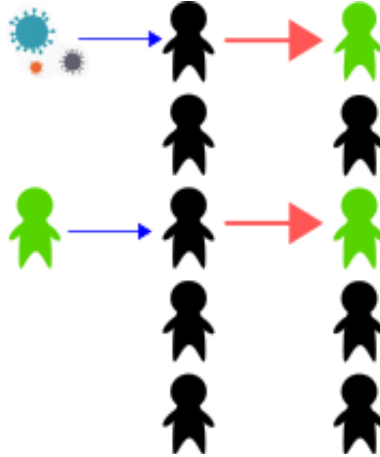


Figure 1: Representation of a self-exciting phenomenon. Individuals can get infected by either direct contact with a virus (spontaneous occurrence) or contact with other infected individuals (self-excitation).

When studying such a process, it is natural to keep a record of the time-stamps when all the events took place. In order to make inferences from these data sets, we need a mathematical model that allows us to understand the occurrence mechanism behind a given event series. In this work, we make use of the model proposed by Alan Hawkes in 1971 [12].

The Hawkes Process is a point process in which future event occurrences are influenced by the history of previous events. It is characterized by the rate of event-occurrence $\lambda(t)$ which comprises both external and internal causes. External influences are represented as a background rate of spontaneous occurrence ρ . Meanwhile, internal self-exciting causes are described by a reproduction number α , which indicates the fraction of additional events generated by one event, and a weighted sum over all the previous events t_k . The weights for the sum are given by a *kernel* function $h(t)$ which indicates how the influence of a past event changes in time. The occurrence rate $\lambda(t)$ takes the form:

$$\lambda(t) = \rho + \alpha \sum_{t_k < t} h(t - t_i)$$

Since a particular Hawkes process is completely determined by the occurrence rate $\lambda(t)$, it is possible to take some fixed choice of the parameters $(\rho, \alpha, h(t))$ and obtain an event series that follows that particular process. One could, however, take the inverse approach: obtain a collection of time-stamps t_i from the observation of some phenomenon and recover, from only that event series, an estimation of the parameters $(\hat{\rho}, \hat{\alpha}, \hat{h}(t))$.

Estimation using point processes has been studied before; some methods involve the use of Spectral Analysis with the manipulation of Fourier transforms [7], Machine Learning algorithms [28], or the solution of Wiener-Hopf integral equations of the second kind [5]. All of these methods make use of the first and second order statistics of the Hawkes process; namely, the *average firing rate* $\bar{\lambda}$ and the *covariance density function* $\phi(t)$.

In this work, we made use of the lattermost approach for estimating the reproduction number $\hat{\alpha}$ and the kernel function $\hat{h}(t)$. For this, we suggested a non-parametric method for computing an estimate of the covariance density and implemented a Reproducing Kernel Hilbert Space method for solving the Wiener-Hopf integral equation. We then compared our method to other existing methods and concluded that ours is superior in both computation speed and accuracy of estimation. In doing so, not only did we consider the error resulting from numerical computations, but also the noise that is inherent to stochastic phenomena.

Thesis objectives and structure

The objective of this work is to suggest a method for estimating the parameters of the Hawkes Process for a given spike train and to express them as a set of real numbers and a function of time. This method involves the construction of histograms for estimating the autocorrelation function of a given spike train, as well as a way for selecting the histogram bin size. Then, an integral equation is solved for the estimation of the parameters of the process. The method is tested with simulated data, and compared in performance against other methods. Finally, the method is applied to real data taken from the brain of a monkey.

In section 2, we present the theoretical background of the Hawkes process and the conventions that we will consider in this work. In section 3, we introduce the theory for our estimation method of the kernel function and our two approaches for the estimation of the reproduction number. We also provide rules for selecting the optimal parameters for our methods. Section 4 is dedicated for the analysis of performance of our method on simulated data. We present the algorithm used for generating the data, and we prove that the simulated spike trains follow indeed a Hawkes process. In addition, we compare the performance against existing methods. Section 5 contains the result of applying our method to biological data. The data sets used come from the neuronal firings in the brain of a monkey. Finally, in section 6, we present our concluding remarks and a discussion about the viability of our method on real-life situations. Some technical (but important) derivations about our estimation methods are left for the appendix.

2 The Hawkes Process

Before going into detail about the estimation methods that are used in this work, an introduction to the mathematics of the Hawkes Process is presented.

2.1 Theoretical frame

The Hawkes Process can be studied as a counting process, or as a point process. These concepts, as presented in [14] are defined as below:

Definition 1 (Counting Process). A counting process is a stochastic process $N(t)$ which for every time t it takes values in the non-negative integers, and represents the number of events that have occurred up to time t . $N(t)$ is a step function, continuous from the right, and whose jumps are unit increments.

A counting process is understood as the total number of events that have occurred in time, up to time t . It is possible, however, to have a conjugate interpretation of this process by thinking of the individual arrival times as a sequence of ordered random variables that have the time-stamps of the events as an output. This is precisely what a point process is.

Definition 2 (Point Process). A point process is a sequence of non-negative random variables $\mathbf{T} = \{T_1, T_2, T_3, \dots\}$ such that $\mathbb{P}(0 \leq T_1 \leq T_2 \leq \dots)$ and that does not *explode*, i.e. the number of random variables in a bounded region is finite with probability one.

It is important to note that often in literature definitions 1 and 2 are used interchangeably.

Definition 3 (Spike Train). A realization of a point process is a sequence of real numbers $\mathbb{S} = \{t_1, t_2, \dots | t_1 \leq t_2 \leq \dots\}$. Such a sequence is called a *spike train* and represents the times at which events took place. When treating with data, one usually has a finite sequence $\mathbb{S} = \{t_1, t_2, \dots, t_n\}$. In this case, we say that we have a spike train of length $T = t_n$.

With these definitions, it is now possible to introduce the Hawkes Process.

Definition 4 (Hawkes Process). A Hawkes Process is a point process in which the rate of event-occurrence $\lambda(t)$ is given by

$$\frac{\mathbb{E}\{dN(t)|H_t\}}{dt} \equiv \lambda(t) = \rho + \alpha \int_{-\infty}^t h(t-u)dN(u), \quad (1)$$

where H_t is the history of events up to time t , $\rho > 0$ is the baseline rate, $\alpha \geq 0$ is the reproduction number, and $h(t)$ is the self-exciting kernel function that satisfies $h(t) = 0$ for $t < 0$ and $\int_0^\infty h(t)dt = 1$. Hawkes originally required that $h(t) \geq 0$ [12], but it has been shown [8] that the kernel can be allowed to take negative values. Together with this, it is also required that the process is stationary and that no more than one spike can occur at the same time. Strictly speaking, $\frac{\mathbb{E}\{dN(t)|H_t\}}{dt}$ should be understood as $\lim_{\Delta t \downarrow 0} \frac{\mathbb{E}\{N(t+\Delta t) - N(t)|H_t\}}{\Delta t}$ whenever the limit exists.

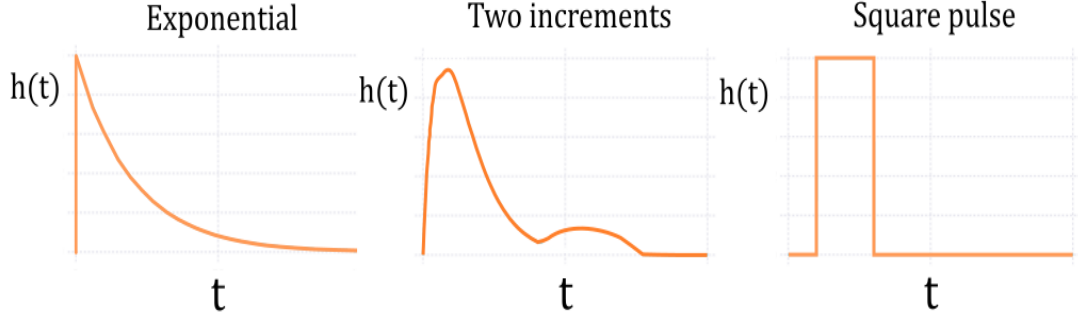


Figure 2: Some examples of self-exciting kernels. The kernel function indicates how the influence of one event propagates to future times.

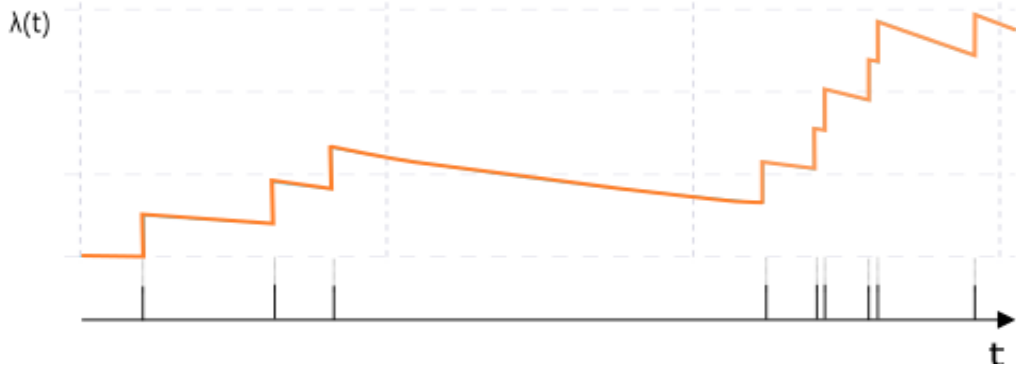


Figure 3: Example of an occurrence rate function with an exponential kernel. Each time a new event occurs, a ‘copy’ of the kernel function $h(t)$ is added, scaled by the reproduction number α .

2.2 First and second order statistics

The first thing to note from the definition of the Hawkes process is that the measure $dN(u)$ can be written as a sum of Dirac’s delta functions centered at the time of each event

$$dN(u) = \sum_k \delta(u - t_k) du. \quad (2)$$

By doing this, the firing rate $\lambda(t)$ can be rewritten in a more intuitive way

$$\lambda(t) = \rho + \alpha \sum_{t_k < t} h(t - t_k), \quad (3)$$

so that the sum is taken over all the times t_k where an event took place, up to time t . An illustration of this can be seen in figure 2. We can see that if $\alpha = 0$, then the process is an homogeneous Poisson process, in which the average number of arrivals is constant in time $\lambda(t) = \rho$. In this case, the average firing rate is $\mathbb{E}(\lambda(t)) = \rho$. For the Hawkes Process, the average firing rate can be computed by making use of the stationarity:

$$\bar{\lambda} \equiv \mathbb{E}(\lambda(t)) = \mathbb{E} \left(\rho + \alpha \int_{-\infty}^t h(t - u) dN(u) \right) = \rho + \alpha \int_{-\infty}^t h(t - u) \mathbb{E}(dN(u)). \quad (4)$$

Because of stationarity $\mathbb{E}(dN(u)) = \bar{\lambda}$ so

$$\bar{\lambda} = \rho + \bar{\lambda}\alpha \int_0^\infty h(u)du = \rho + \bar{\lambda}\alpha, \quad (5)$$

and we conclude that

$$\bar{\lambda} = \frac{\rho}{1 - \alpha}. \quad (6)$$

From here, it is possible to see that in order for the process to not explode, then $\alpha < 1$. The next quantity of interest for estimation is the autocovariance function. The importance of this quantity will become clear with the following Theorem:

Theorem 1. Covariance density of the Hawkes process Let $N(t)$ be the counting process associated to a Hawkes process, then its covariance density function $\phi(t)$ will satisfy the integral equation

$$\phi(t) = \alpha\bar{\lambda}h(t) + \alpha \int_{-\infty}^t h(t-v)\phi(v)dv.$$

Proof. Consider the *covariance density* defined by

$$\phi(t) = \mathbb{E} \left\{ \frac{dN(s+t)}{dt} \frac{dN(s)}{ds} \right\} - \bar{\lambda}^2. \quad (7)$$

In literature, this quantity is also referred to as the *autocorrelation function* for the Hawkes process, even though it is more strictly a covariance since it is not normalized. In this work, we will also use these two terms interchangeably. We have that $\phi^{(c)}(t) = \phi^{(c)}(-t)$, but at $t = 0$ there is a singularity, since $\mathbb{E}((dN(s))^2) = \mathbb{E}(dN(s))$ because we only allow jumps of length 0 or 1. To remove this singularity, we define the *complete covariance density*

$$\phi^{(c)}(t) \equiv \phi(t) + \lambda\delta(t). \quad (8)$$

By using (1) to replace $dN(s+t)$ in (7) we obtain for $t > 0$

$$\phi(t) = \mathbb{E} \left\{ \frac{dN(s)}{ds} \left(\rho + \alpha \int_{-\infty}^{s+t} h(s+t-u)dN(u) \right) \right\} - \bar{\lambda}^2. \quad (9)$$

If we note that $\rho\bar{\lambda} - \bar{\lambda}^2 = -\alpha\bar{\lambda}^2$ and make the change of variable $u = v + s$ we get

$$\alpha \int_{-\infty}^t h(t-v)\mathbb{E} \left\{ \frac{dN(s)}{ds} \frac{dN(s+v)}{dv} \right\} dv - \alpha\bar{\lambda}^2, \quad (10)$$

and since $\int_{-\infty}^t h(t-v)dv = 1$, then

$$\phi(t) = \alpha \int_{-\infty}^t h(t-v) \left(\mathbb{E} \left\{ \frac{dN(s)}{ds} \frac{dN(s+v)}{dv} \right\} - \bar{\lambda}^2 \right) dv. \quad (11)$$

The term in parenthesis corresponds once again to the definition of the covariance density, however, since $t > 0$, it is necessary to avoid the singularity:

$$\phi(t) = \alpha \int_{-\infty}^t h(t-v)\phi^{(c)}(v)dv, \quad (12)$$

which from (8) we can write as

$$\phi(t) = \alpha\bar{\lambda}h(t) + \alpha \int_{-\infty}^t h(t-v)\phi(v)dv. \quad (13)$$

□

The expression (13) is an integral equation satisfied by the covariance density, and is key in point process estimation. In order to extract the self-exciting kernel from this expression, it is necessary to first get an estimate of the covariance density from a given random sample and then to solve the integral equation for $h(t)$. The methods for carrying this out are described in the following sections.

3 Estimation

When estimating the parameters of the Hawkes process, especially the reproduction number and the kernel function, there are different approaches that could be followed. One of these strategies involves assuming that the kernel function belongs to a family of functions $\mathcal{F} = \{f(x; \gamma) | \gamma \in \mathbb{R}^n\}$ where f is a fixed function and depends on a set of parameters γ . Then, the estimation method revolves around finding the parameter $\gamma^* \in \mathbb{R}^n$ such that $f(x; \gamma^*)$ best estimates the kernel function. Because of this nature, this approach is called *parametric estimation*, and is the most useful when there is enough information to make a reasonable guess of the family of functions $\mathcal{F}$ that will be considered. One common choice is a family of exponential functions $\mathcal{F} = \{\frac{1}{\tau}e^{-\frac{t}{\tau}} | \tau \in \mathbb{R}\}$ as in [16].

If there is no previous knowledge about the nature of the kernel function, then a second approach known as *non-parametric estimation* can be used. In this case, an estimate of the overall shape of the kernel function is obtained. It is important to notice that parametric estimation methods have to deal with less degrees of freedom, and thus, if the initial guess was reasonable, then the estimation error will be smaller in general. However, if that initial guess was far from the real kernel function, then the results can deviate considerably. In this work, the focus will be non-parametric estimation, and no comparison with parametric estimation methods will be made.

3.1 Estimation of the mean firing rate

The estimation of the mean firing rate $\bar{\lambda}$ is straightforward. Given a spike train $\mathbb{S}$ of length T , let N_T be the number of events in the spike train. Then, the estimator

$$\hat{\bar{\lambda}} = \frac{N_T}{T}, \quad (14)$$

converges to the value of $\bar{\lambda}$ for $T \rightarrow \infty$, as was shown in [4]. Since equation (6) relates $\bar{\lambda}$ with the constants α and ρ , by getting estimates of $\bar{\lambda}$ and α , we could get an estimate of the baseline rate ρ as well.

3.2 Estimation of the autocorrelation function

A necessary step in the estimation of the self-exciting kernel is to find an estimate of the covariance density in order to solve (13). Density estimation has been studied before, and there is an existing collection of methods presented in previous works along with the analysis of their error. Some examples we will discuss briefly are the Parzen window [21] and the creation of bar histograms [25].

3.2.1 Existing methods

In the Parzen window method, for given a *random sample*, i.e. a set of identically distributed independent random variables, $X_1, X_2, \dots, X_N$ with absolutely continuous distribution function $F(x) = \mathbb{P}(X \leq x)$ and density function $f(x)$ such that

$$F(x) = \int_{-\infty}^x f(t)dt,$$

then, the density function $f(x)$ can be estimated by the quantity

$$\hat{f}(x) = \frac{1}{Nh} \sum_{j=1}^N K\left(\frac{x - X_j}{\Delta}\right), \quad (15)$$

where $\Delta > 0$ and $K(y)$ a bounded *window function* satisfying $\int_{-\infty}^{\infty} |K(y)| dy < \infty$ (it is often asked that $\forall y, K(y) > 0$ and $\int_{-\infty}^{\infty} K(y) dy = 1$ for convenience). It can be seen that once the function $K(y)$ is fixed, the parameter Δ known as the *bandwidth* or *bin size* can be arbitrarily chosen. This turns out to be a similarity among the density estimation methods, where by changing the value of Δ , the quality of the estimation gets modified. The task is now to find the value Δ^* that produces the best estimate of the density function f . As presented in [21], the theoretical optimal value for Δ has a power law asymptotic behavior that scales like $N^{-\frac{1}{2r+1}}$, and the Mean Integrated Squared Error (MISE) scales as the power law $N^{-\frac{2r}{2r+1}}$, where r is called the *characteristic exponent* of the window function, and is defined as the first integer such that the limit $\lim_{u \rightarrow 0} (1 - \tilde{K}(u))/|u|^r$ is different than zero and $\tilde{K}(u)$ is the Fourier transform of the window function.

For the creation of bar histograms, given the random sample $X_1, X_2, \dots, X_N$ from a continuous probability density function f with two continuous and bounded derivatives, we select a bin size Δ , and divide the real number line up to some number T in intervals (or bins) Δx_i of length Δ so that each element of the sample falls into one of these bins. We then count the amount of elements in each bin $v_i(x)$ and draw a bar of height

$$\hat{f}(x) = \frac{v_i(x)}{Nh}. \quad (16)$$

The resulting histogram will have a shape that approximates the shape of the underlying density function f . Once again, the quality of the estimation relies on the choice of the bandwidth Δ . In [25] it is shown that the optimal bandwidth scales with N as a power law $N^{-\frac{1}{3}}$ and the MISE scales as $N^{-\frac{2}{3}}$.

3.2.2 Proposed method of estimation

The purpose is to obtain an estimator $\hat{f}(x)$ for an unknown probability density $f(x)$, given a random sample $X_1, X_2, \dots, X_N$. For this, we start by constructing a bar histogram using a bin size Δ , and then we use the value at the midpoints of each bin to build a histogram using *nth-order splines*, which are polynomial interpolations defined by parts. As with the previous methods, there are many possibilities for the choice of Δ , but there is an optimal bin size Δ^* for which the histogram more closely resembles the true distribution. It will be shown how this optimal bin size scales with the size of the random sample and the order of the spline.

The criterion that will be used to analyze the quality of our estimator will be the minimization of the MISE. To compute the MISE, we first need to consider the MSE at each point x and then integrate it over the interval of interest. The Mean Squared Error at point x is given by

$$MSE[\hat{f}](x) = Var(\hat{f}(x)) + b^2(x), \quad (17)$$

where $Var(\hat{f}(x))$ is the variance of the estimator and $b^2(x)$ is the bias squared. This representation of the error as a sum of two terms is useful in this case because it allows

to analyze each term independently. The variance of the estimator will come from the variability of the histogram heights. Once these heights are determined, the bias of the estimator will be entirely due to the order of the polynomial approximation we use, since the points which the spline will go through will be the same for any chosen order. Let us then compute each of these terms separately.

Proposition 1. (Variance of $\hat{f}(x)$). Let $X_1, X_2, \dots, X_N$ be a random sample from a distribution with an unknown probability density function $f(x)$ that has two continuous bounded derivatives (as a result f will be absolutely continuous). Let $\{t_i\}_{i=0}^M$ the bin endpoints of a bar histogram $\hat{f}(x)$ over the interval of interest $[t_0, T]$, where all the t_i are equally spaced a distance $\Delta > 0$ and $T \equiv t_n$. Then, the variance for this estimator will be

$$\text{Var}\{\hat{f}(x)\} = \frac{f(x)}{Nh} + O(1/N). \quad (18)$$

Proof. To build the bar histogram, we count the number of elements from the sample that fall in each bin, and then we assign a height to each bar that corresponds to the count in each bin; normalized by the bin size and the sample size. Let $x \in [t_1, t_M]$, then the value of our bar histogram estimator at point x is

$$\hat{f}(x) = \frac{\gamma_i(x)}{Nh} \mathbb{I}_{[t_i, t_i + \Delta)}(x), \quad (19)$$

where $\mathbb{I}_{[t_i, t_i + \Delta)}(x)$ is the characteristic function of the bin $[t_i, t_i + \Delta)$ and $\gamma_i(x)$ is the number of elements in that bin. The probability that one element of the sample is in bin i is given by $\mathbb{P}(X_j \in [t_i, t_i + \Delta)) = \int_{t_i}^{t_i + \Delta} f(t) dt \equiv p_i$, and corresponds to a Bernoulli trial. Thus, the number of elements in each bin $\gamma_i(x)$ has a binomial distribution $\text{Bin}(N, p_i)$. Since the variance of a binomial distribution $\text{Bin}(n, p)$ is given by $np(1 - p)$, so we have that

$$\text{Var}(\hat{f}(x)) = \text{Var}\left(\frac{\gamma_i(x)}{Nh}\right) = \frac{\text{Var}(\gamma_i(x))}{N^2 h^2} = \frac{N p_i (1 - p_i)}{N^2 h^2} = \frac{p_i (1 - p_i)}{N h^2}. \quad (20)$$

Now, by expanding $f(t)$ around point x using its Taylor series up to first order we get

$$f(t) = f(x) + f'(x)(t - x) + O(\Delta^2). \quad (21)$$

By using (21) to approximate p_i we obtain

$$p_i = \int_{t_i}^{t_i + \Delta} (f(x) + f'(x)(t - x) + O(\Delta^2)) dt = h f(x) + \frac{1}{2} f'(x) (\Delta^2 - 2h(x - t_i)) + O(\Delta^3).$$

Notice now that in the product $p_i(1 - p_i)$ there is only one term that is linear in Δ :

$$\frac{p_i(1 - p_i)}{N h^2} = \frac{(h f(x) + f'(x) O(\Delta^2))(1 - h f(x) + f'(x) O(\Delta^2))}{N h^2} = \frac{h f(x) + O(\Delta^2)}{N h^2},$$

and hence

$$\frac{p_i(1 - p_i)}{N h^2} = \frac{f(x)}{N h} + \frac{O(\Delta^2)}{N h^2} = \frac{f(x)}{N h} + \frac{O(1)}{N},$$

thus

$$\text{Var}\{\hat{f}(x)\} = \frac{f(x)}{N h} + O(1/N), \quad (22)$$

which is the result we were looking for. □

Now, we want to find the bias of our estimator. The bias will come from the error of approximating the true distribution by spline functions, where the interpolation nodes are already fixed. This approximation by splines is done in a similar way to Lagrange interpolation methods that can be consulted in [13].

Proposition 2. (Bias of $\hat{f}(x)$). Let $f : [a, b] \rightarrow \mathbb{R}$ a function that has $r + 1$ continuous bounded derivatives. Let $p_r(x)$ be the r -th order polynomial approximation of f that has the same value as f in $r + 1$ equally spaced nodes $\{x_0, x_1, \dots, x_r\}$ separated by a distance $\Delta = \frac{b-a}{r}$. Then, the upper bound for the estimation error will be given by

$$\max |f(x) - p_r(x)| \leq \Delta^{r+1} \frac{1}{4(r+1)} \max |f^{(r+1)}(z_0)|, \quad (23)$$

for some $z_0 \in [a, b]$.

Proof. The premise of approximation with splines is that the approximation is exact at every node. The nodes $\{x_0, x_1, \dots, x_r\}$ for the interpolation in this case will be the midpoints of the bar histogram; we need to notice that if we have a spline of order r , then we will need $r + 1$ nodes for constructing it. Because of this restriction, consider the following polynomial

$$w_{r+1}(x) \equiv (x - x_0)(x - x_1) \dots (x - x_r) = \prod_{j=0}^r (x - x_j), \quad (24)$$

where the x_i are the nodes for the splines. It is clear that this polynomial is zero at every single node.

Since the error is zero at every node, we assume that x does not equal to any node and we consider $p_r(x)$ as the polynomial interpolate (spline) at the point x . Now, we consider the following function:

$$F(t) = f(t) - p_r(t) - cw_{r+1}(t),$$

and

$$c = \frac{f(x) - p_r(x)}{w_{r+1}(x)}.$$

We assumed that x is not a node, so c is well defined. Since $p_r(x_i) = f(x_i)$ for every single node, then:

$$\begin{aligned} F(x_i) &= 0, \quad i = 1, \dots, r, \\ F(x) &= 0. \end{aligned}$$

The nodes are different and also x is different from the nodes, so this function has at least $r + 2$ different zeros in the interval of length rh . By the Mean Value Theorem, we know that between any two roots of F , there should be a root of F' . So then F' has at least $r + 1$ different zeros, and F'' has r different zeros, and so on, all the way down to $F^{(r+1)}$ which must have one zero. Let z_0 be one zero of $F^{(r+1)}(t)$. We have

$$0 = F^{(r+1)}(z_0) = f^{(r+1)}(z_0) - c(r+1)! = f^{(r+1)}(z_0) - (r+1)! \frac{f(x) - p_r(x)}{w_{r+1}(x)}.$$

Notice that the $(r+1)$ -th derivative of $w_{r+1}(t)$ is precisely $(r+1)!$, so the result follows:

$$f(x) - p_r(x) = \frac{w_{r+1}(x)}{(r+1)!} f^{(r+1)}(z_0). \quad (25)$$

We can now take absolute value and maximize the right hand side to obtain a bound for the error.

$$|f(x) - p_r(x)| \leq \max \left| \frac{w_{r+1}(x)}{(r+1)!} f^{(r+1)}(z_0) \right|. \quad (26)$$

We can bound this even further by using the fact that the nodes are equally spaced at a distance Δ . We first notice that the problem reduces to find a bound for

$$\max |w_{r+1}(x)| = \max \left| \prod_{j=0}^r (x - x_j) \right|,$$

which, as shown in Appendix A, can be bound in the following way

$$\max \left| \prod_{j=0}^r (x - x_j) \right| \leq \Delta^{r+1} \frac{1}{4} r!, \quad (27)$$

and the result follows:

$$|f(x) - p_r(x)| \leq \Delta^{r+1} \frac{1}{4(r+1)} \max |f^{(r+1)}(z_0)|.$$

□

With this result we can finally compute the MISE for our estimation method. This is done in the following Proposition.

Proposition 3. (Asymptotic behavior of the MISE and optimal bin size). Let $X_1, X_2, \dots, X_N$ be a random sample from a distribution with an unknown probability density function $f(x)$ that has $r+1$ continuous bounded derivatives. Let $\{x_i\}_{i=0}^M$ the mid-points of a bar histogram constructed as in Proposition 1. If $\hat{f}(x)$ is an r -th order polynomial spline interpolation of f that passes through all the points $\{x_i\}_{i=0}^M$, then there is an *optimal* value Δ^* of the bin-size that minimizes the MISE, and in this case, Δ^* and the MISE will have the following asymptotic scaling laws:

$$\Delta^* \propto N^{-\frac{1}{2r+3}}, \quad MISE[\hat{f}] \propto N^{-\frac{2r+2}{2r+3}}. \quad (28)$$

Proof. We start by using (17) along with (18) and (23) to find the MSE at each point x .

$$MSE[\hat{f}](x) = \frac{f(x)}{Nh} + \Delta^{2r+2} \frac{1}{16(r+1)^2} \sum_i \max |f^{(r+1)}(z_{0_i})|^2 \mathbb{I}_{[t_i, t_i+rh)}(x).$$

It is important to mention that the point z_{0_i} in (23) is given for each set of $r+1$ nodes independently. The sum term comes from this fact and indicates that according to which set of nodes x is in, the choice of the z_{0_i} will change accordingly. If we now integrate over the interval of estimation, considering that the density is normalized, then we have that:

$$MISE[\hat{f}] = \frac{B(f)}{Nh} + \frac{1}{16(r+1)^2} C_r(f) \Delta^{2r+2}, \quad (29)$$

where

$$B(f) = \int_{t_0}^T f(t)dt, \quad C_r(f) = \sum_i \int_{t_0}^T \max |f^{(r+1)}(z_{0_i})|^2 \mathbb{I}_{[t_i, t_i+rh)}(t)dt,$$

are constants that depend on f , the interval of interest, and the order of the spline. We now differentiate this expression to find the optimal bin size:

$$0 = -\frac{B(f)}{Nh^2} + \frac{1}{16(r+1)^2} C_r(f)(2r+2)\Delta^{2r+1} = -\frac{B(f)}{Nh^2} + \frac{C_r(f)}{8(r+1)} \Delta^{2r+1}.$$

From this, we find the scaling of the optimal bin size:

$$\Delta^* = D_r(f)N^{-\frac{1}{2r+3}}, \quad (30)$$

where $D_r(f) = \left(\frac{8(r+1)B(f)}{C_r(f)}\right)^{-\frac{1}{2r+3}}$ is a constant, N is the size of the sample, and r is the order of the splines. Now we can substitute this expression for Δ^* into (29) to find that

$$MISE[\hat{f}] = \frac{B(f)}{D_r(f)} N^{-\frac{2r+2}{2r+3}} + \frac{C_r(f)D_r(f)^{2r+2}}{16(r+1)^2} N^{-\frac{2r+2}{2r+3}},$$

which can be expressed as

$$MISE[\hat{f}] = K_r(f)N^{-\frac{2r+2}{2r+3}}, \quad (31)$$

where $K_r(f) = \frac{B(f)}{D_r(f)} + \frac{C_r(f)D_r(f)^{2r+2}}{16(r+1)^2}$ is a constant that depends on f and the order of the splines r . □

We have shown that the optimal bin-size and estimation MISE decrease as we increase the size of the sample N . Furthermore, we have explicitly indicated the theoretical asymptotic scaling laws of this decrease. It is possible to compare the asymptotic behavior of our estimator for $r = 0$ which corresponds to 0-th order splines (constant polynomials). This corresponds to just a bar histogram. In our case, the scalings go as $\Delta^* \propto N^{-\frac{1}{3}}$ and $MISE[\hat{f}] \propto N^{-\frac{2}{3}}$ which are consistent with the scaling found by Scott [25] with his estimator (16). In addition, the scaling laws of our estimators are consistent with those presented in [18, 26], which correspond to the maximum asymptotic convergence speeds for non-parametric estimators of densities with r continuous bounded derivatives.

In the figure below, our method is showcased when applied to a random sample from the $Gamma(2, 3)$ distribution.

$$f(x) = \frac{1}{9}xe^{-\frac{x}{3}}\mathbb{I}_{[0,\infty)}(x).$$

Showcase of our density estimation method.

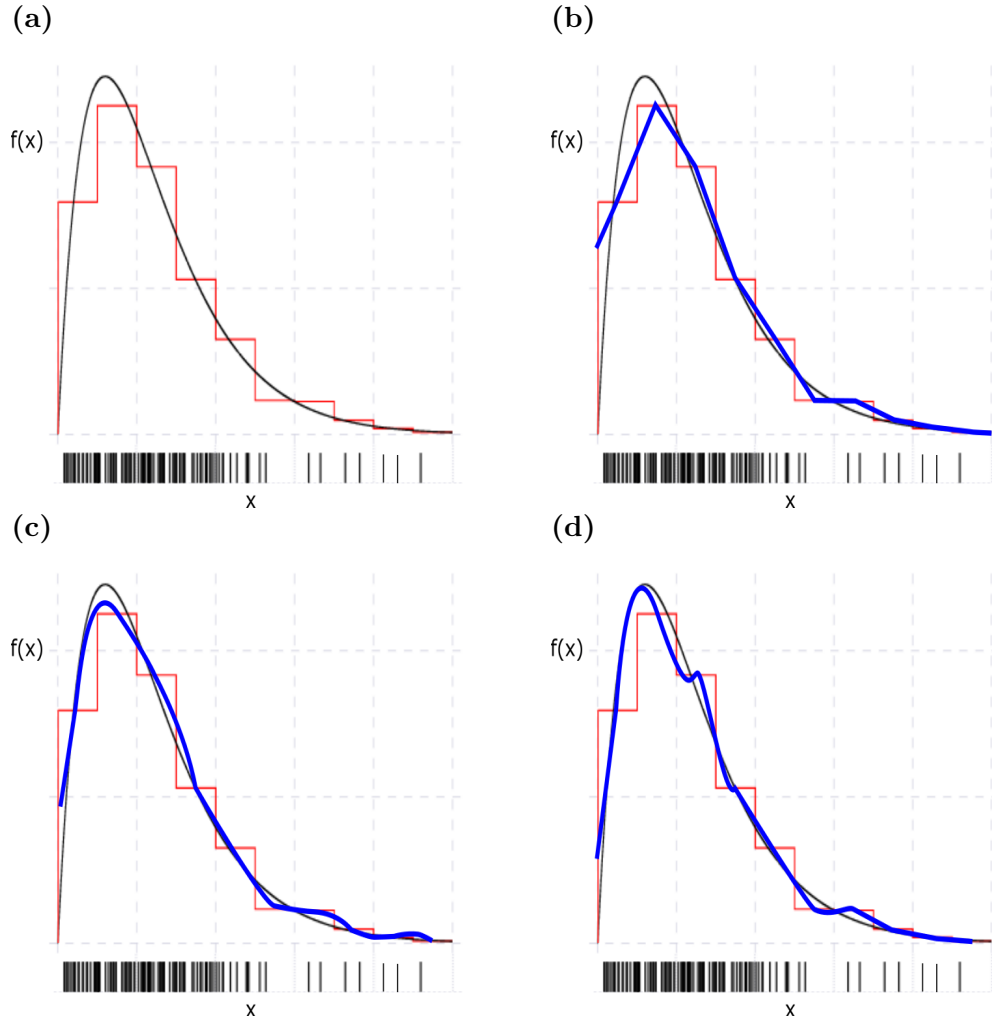


Figure 4: Our density estimation method applied to a random sample of a Gamma(2,3) distribution with splines of: **(a)** Zeroth order **(b)** First order **(c)** Second order **(d)** Third order.

3.2.3 Estimation of $\phi(t)$ given a spike train

The previous results were obtained for an arbitrary density $f(t)$, so they can be used for the particular case of the covariance density $\phi(t)$. Because of the symmetry of ϕ , it is possible to only estimate it on an interval $[0, T]$, and then consider that $\phi(t) = \phi(-t)$ for the interval $[-T, 0)$.

The autocorrelation function $\phi(t)$ of a spike train $\mathbb{S}$ can be understood as a measure of the amount of intervals of length t between spikes [10]. This can be interpreted as a density; although it is not necessarily normalized to unity, we can use non-parametric density estimation techniques to find a good approximation of it. Available data will come in the form of a spike train $\mathbb{S} = \{t_1, t_2, \dots, t_N\}$ of length N . It is necessary to first treat these data in order to apply our method. First, we construct the collection

$$\mathbb{S}' = \{s_{ij}\}_{1 \leq j \leq i \leq N},$$

where the s_{ij} are given by

$$s_{ij} \equiv t_i - t_j, \quad t_i, t_j \in \mathbb{S}, \quad 1 \leq j \leq i \leq N. \quad (32)$$

That is, the collection $\mathbb{S}'$ contains all lengths of the positive intervals between spikes in $\mathbb{S}$. It is this collection $\mathbb{S}'$ to which the methods for density estimation are applied to. This will yield an estimation of $\phi(t) + \bar{\lambda}$, so we just need to subtract the estimated average rate computed using (14). This process can be seen in the figure below.

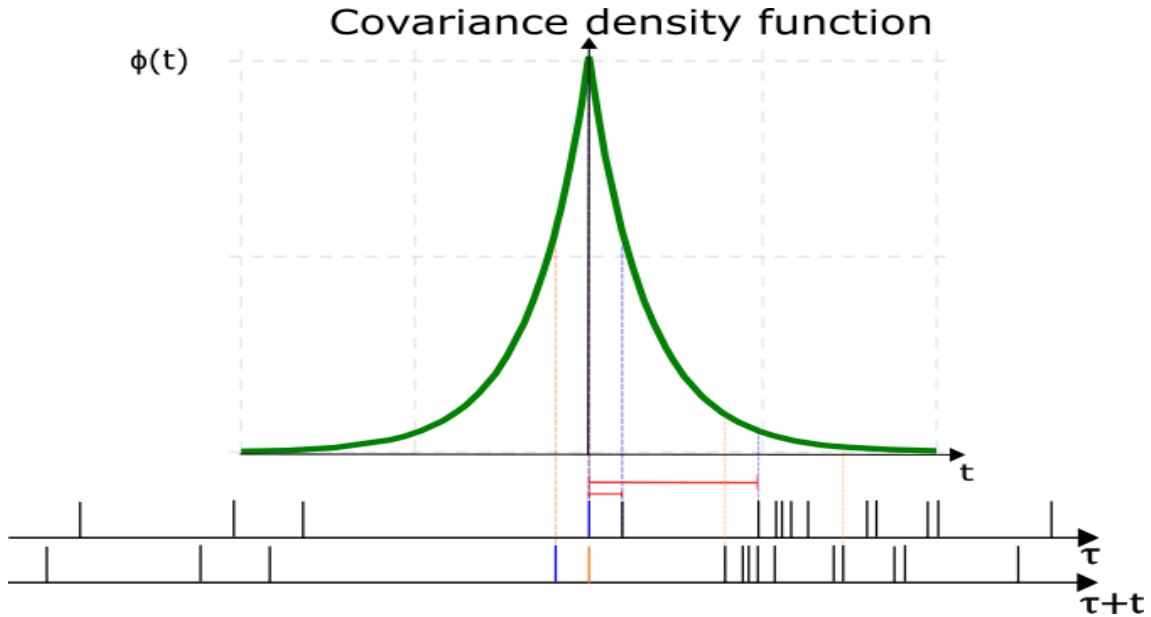


Figure 5: Construction of the autocorrelation function from a spike train. We count the fraction of intervals of length t from a reference spike and shift the reference spike until we sweep the whole event series.

As mentioned before, the quality of the estimation depends on the choice of the bandwidth Δ . We have shown, theoretically, the value of the optimal choice; however, we have no knowledge of the original distribution that we are trying to estimate. We discuss next the issue of choosing the optimal bandwidth parameter.

3.3 Selection of the optimal Δ^*

In [6] the authors implement a cross-validation approach to select the optimal bandwidth for estimating the covariance density with the Parzen window method. In our case, we will follow steps similar to what Scott did in [25] and use the insight gained from knowing the scaling laws of the optimal bin-size (28) to write an empirical law for selecting Δ^* for any given spike train. Our suggestion is that such an empirical law, in the case of the Hawkes process, should have the following structure:

$$\Delta^* = C_r \sigma N^{-\frac{1}{2r+3}}, \quad (33)$$

where r is the order of the splines, σ is the width of the covariance density, and C_r is a constant that depends on the order of the spline being used. In practice, we do not know the value of σ because the original density is not known. For this reason, we suggest to normalize the covariance density and use its standard deviation as an estimate $\hat{\sigma}$ of the width.

As it was mentioned previously, if we apply the density estimation method to the interval lengths s_{ij} , we get an estimation of $\phi(t) + \bar{\lambda}$. For this reason, to find $\hat{\sigma}$ it is necessary to subtract the average rate, and also, to include a normalization constant. The normalization constant, for a long enough time W , will be given by

$$\int_0^W (\phi(t) + \bar{\lambda}) dt \approx \sum_{i=1}^N \sum_{j=1}^i \mathbb{I}_{[0,W]}(s_{ij}), \quad (34)$$

$$A(W) = \sum_{i=1}^N \sum_{j=1}^i \mathbb{I}_{[0,W]}(s_{ij}) - \hat{\lambda}W, \quad (35)$$

thus, the estimation of the width is

$$\int_0^W t^2 (\phi(t) + \bar{\lambda}) dt \approx \sum_{i=1}^N \sum_{j=1}^i s_{ij}^2 \mathbb{I}_{[0,W]}(s_{ij}), \quad (36)$$

$$\hat{\sigma} = \frac{\int_0^W t^2 \phi(t) dt}{A(W)}, \quad (37)$$

$$\hat{\sigma} = \frac{1}{A(W)} \left(\sum_{i=1}^N \sum_{j=1}^i s_{ij}^2 \mathbb{I}_{[0,W]}(s_{ij}) - \hat{\lambda} \frac{W^3}{3} \right). \quad (38)$$

With expression (38), by knowing the estimate $\hat{\sigma}$ and the sample size N , it is possible to compute the optimal bin size with the equation

$$\Delta^* = C_r \hat{\sigma} N^{-\frac{1}{2r+3}} \quad (39)$$

given that we have fixed the values of the constants C_r . Their values will be obtained experimentally for each spline order with simulated data.

Now that we have a method for computing an estimate of $\phi(t)$, we need a method to solve the integral equation (13). For this, let us apply a change of variable and define the following functions.

$$u \rightarrow t - v, \quad du \rightarrow -dv, \quad \Phi(t) \equiv \frac{\phi(t)}{\bar{\lambda}}, \quad f(t) \equiv \alpha h(t).$$

So equation (13) becomes

$$\Phi(t) = f(t) + \int_0^\infty \Phi(t-u)f(u)du. \quad (40)$$

Equation (40) is known as a Wiener-Hopf integral equation of the second kind. Here the constant α has been included into the new kernel function $f(t)$, but since $h(t)$ is normalized, an estimate of α can be taken as

$$\hat{\alpha} = \int_0^\infty f(t)dt. \quad (41)$$

Alternatively, we later suggest a method for estimating α independently. We have shown that the information about the parameters of the Hawkes process can be obtained by estimating the kernel function $f(t)$. We focus on this subject now.

3.4 Estimation of the kernel function

Here, two methods for solving (40) are presented. In a subsequent section, a comparison of their performance is made. It has already been pointed out that the function $\phi(t)$, and consequently $\Phi(t)$, has a singularity at $t = 0$. If we want to solve the integral equation for $t \geq 0$, then we need to use methods that allow to address this problem. This will be the case of the *Nyström method*, which was previously used in [6], and the *Reproducing Kernel Hilbert Space method* (RKHS), which we suggest in this work for the purpose of kernel estimation.

3.4.1 The Nyström method

This is a conventional method for solving many kinds of integral equations, and can be modified accordingly to solve (54). The Nyström method, as presented in [22], focuses on solving integral equations of the form

$$g(t) = y(t) - \sigma \int_a^b k(t-s)y(s)ds, \quad (42)$$

where $\sigma \in \mathbb{R}$, $k(t) \in L_1(\mathbb{R})$ with r continuous derivatives for some integer r , $g(t) \in L_p[0, \infty)$, for $(0 \leq p < \infty)$, and $y(t)$ is the solution to the integral equation. The idea of this method is to approximate the integral term by a *gaussian quadrature* rule. With this, it is possible to reduce the problem to solving a linear system of equations.

The theory of gaussian quadratures is closely related to the theory of orthogonal polynomials. In this work, specific details are not mentioned, but the main result that we rely on is presented as a definition.

Definition 5 (Gauss-Legendre Quadrature Rule). Let $f : [-1, 1] \rightarrow \mathbb{R}$ an integrable function, then

$$\int_{-1}^1 f(x)dx \approx \sum_{j=1}^N w_j f(x_j), \quad (43)$$

where x_j are the zeros of the N -th order Legendre polynomial $P_N(x)$ in the interval $[-1, 1]$ and $w_j > 0$ are called the weights, which are given by

$$w_j = \frac{2}{(1-x_j^2)[P_N'(x_j)]^2}. \quad (44)$$

If in a more general way $f : [a, b] \rightarrow \mathbb{R}$ is integrable, then

$$\int_a^b f(x)dx \approx \sum_{j=1}^N w_j^* f(x_j^*), \quad (45)$$

where x_j^* are the zeros of the N -th order Legendre polynomial $P_N(x)$ scaled to the interval $[a, b]$ as $x_j^* = \frac{b+a}{2} + x_j \frac{b-a}{2}$ and w_j^* scaled accordingly $w_j^* = \frac{b-a}{2} w_j$.

For simplicity, x_j and w_j will be written regardless of the interval of integration. Now, it is possible to subtract the singularity in (42) along $t = s$ in the following way

$$\int_a^b k(t-s)y(s)ds = \int_a^b k(t-s)[y(s) - y(t)]ds + \int_a^b k(t-s)y(t)ds,$$

and since the last integral is with respect to s , $y(t)$ can be put outside

$$\int_a^b k(t-s)y(s)ds = \int_a^b k(t-s)[y(s) - y(t)]ds + r(t)y(t), \quad (46)$$

where $r(t) = \int_a^b k(t-s)ds$ can be computed by a quadrature rule:

$$r(t) = \int_0^{t-a} k(u)du - \int_0^{t-b} k(u)du \approx \sum_{j=1}^M w_j k(u_j) - \sum_{j=1}^M p_j w_j k(v_j), \quad (47)$$

which still depends on t since the weights and zeros are computed for $[0, t-a]$ and $[0, t-b]$. Equation (42) becomes

$$g(t) = y(t) - \sigma \int_a^b k(t-s)[y(s) - y(t)]ds - \sigma r(t)y(t). \quad (48)$$

Now, the integral over s can be approximated by a quadrature rule, which yields

$$g(t) = y(t) - \sigma \sum_{j=1}^N w_j k(t-s_j)[y(s_j) - y(t)] - \sigma r(t)y(t), \quad (49)$$

where the quadrature points s_j are the corresponding zeros of the Legendre polynomials and w_j their respective weights. Now, we evaluate equation (49) at the same quadrature points, and we obtain

$$g(t_i) = y(t_i) - \sigma \sum_{\substack{j=1 \\ j \neq i}}^N w_j k(t_i - s_j)[y(s_j) - y(t_i)] - \sigma r(t_i)y(t_i). \quad (50)$$

Notation can be made more compact by defining $k_{ij} \equiv k(t_i - s_j)$, $y_i \equiv y(t_i)$, and similarly for the other terms:

$$g_i = y_i - \sigma \sum_{\substack{j=1 \\ j \neq i}}^N w_j k_{ij} [y_j - y_i] - \sigma r_i y_i. \quad (51)$$

Regrouping terms, this equation becomes

$$\frac{g_i}{\mu_i} = y_i - \sigma \sum_{\substack{j=1 \\ j \neq i}}^N \frac{w_j k_{ij}}{\mu_i} y_j, \quad (52)$$

where $\mu_i = 1 + \sigma \sum_{j=1, j \neq i}^N w_j k_{ij} - \sigma r_i$ does not depend on j . Equation (52) can be written in matrix notation

$$\tilde{\mathbf{g}} = (\mathbb{I} - \sigma \tilde{\mathbf{K}}) \mathbf{y}. \quad (53)$$

where $\mathbb{I}$ is the identity matrix, the components of the vector $\tilde{\mathbf{g}}$ are $\frac{g_i}{\mu_i}$, the components of the matrix $\tilde{\mathbf{K}}$ are $\sum_{j=1, j \neq i}^N \frac{w_j k_{ij}}{\mu_i} y_j$, and the vector $\mathbf{y}$ contains the solution to the integral equation evaluated in the quadrature points y_i . Finally, if we want to obtain the solution evaluated at arbitrary points $\{z_1, z_2, \dots, z_l\}$, Nyström methods suggests that it can be done by using equation (52) as an interpolation rule by evaluating in the points of the grid z_i , instead of the quadrature points t_i .

It is possible then to obtain an estimate of the self-exciting kernel by solving equation (42) with $\sigma = -1$, and $g(t) = k(t) = \Phi(t)$. As for the interval of integration, we can select $[0, L]$ with $L > 0$ the value up to where we want to do the kernel estimation, and think of the kernel as zero for $t > L$. By finding the self-exciting kernel $f(t) \equiv \alpha h(t)$, the estimation step concludes.

3.4.2 The Reproducing Kernel Hilbert Space method

This method was introduced in the paper [1]. It consists of approximating the solution of the Wiener-Hopf integral equation as a truncated infinite series of smooth functions. These smooth functions are constructed through a Gram-Schmidt orthogonalization process with the interior product defined in the RKHS. The detailed construction is explained below.

Given the general Wiener-Hopf equation of the second kind

$$g(t) = y(t) + \int_0^\infty k(t-s)y(s)ds, \quad (54)$$

where $k(t) \in L_1(\mathbb{R})$ with r continuous derivatives for some integer r , and $g(t) \in L_p[0, \infty)$, for $(0 \leq p < \infty)$, the purpose is to find a function $y(t)$ that satisfies it. We check that these hypotheses are met by the functions treated in our estimation; both $g(t)$ and $k(t)$ correspond to the function $\Phi(t)$. Since $\Phi(t)$ is a density, though not normalized, $\Phi(t) \in L_1(\mathbb{R})$, and as a consequence $\Phi(t) \in L_1[0, \infty)$. Also, the density estimation method that we propose requires at least one continuous derivative, so the RKHS method is useful in this case. Before proceeding with the solution of (54), it is necessary to present some preliminaries.

Definition 6 (Reproducing Kernel Hilbert Space). A RKHS is a Hilbert Space H of real-valued functions $f : \mathbb{R} \rightarrow \mathbb{R}$, with a norm $\|\cdot\|_H$ induced by its interior product $\langle \cdot, \cdot \rangle_H$, where for any $x \in \mathbb{R}$, there exists some $C > 0$ such that

$$|f(x)| \leq C \|f\|_H \quad \forall f \in H.$$

In other words, the evaluation functional is bounded $\forall x \in \mathbb{R}$ [2].

One particular property of RKHS that is of interest to this method is the existence of the *Reproducing Kernel* of the Hilbert space (not to be confused with the self-exciting kernel of the Hawkes process). The formal justification of this fact is outside the scope of this work, but the definition is presented below.

Definition 7 (Reproducing Kernel). Let H be a RKHS with interior product $\langle \cdot, \cdot \rangle_H$. For each $x \in \mathbb{R}$, there exists a unique $R_x \in H$ such that

$$f(x) = \langle f, R_x \rangle_H \quad \forall f \in H. \quad (55)$$

Since $R_x \in H$, then for any $y \in \mathbb{R}$, there exists a unique $R_y \in H$ so that

$$R_x(y) = \langle R_x, R_y \rangle_H. \quad (56)$$

The *Reproducing Kernel* of H is the function $R : \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$R(x, y) = \langle R_x, R_y \rangle_H. \quad (57)$$

The task now becomes to find $R(x, y)$ in a Hilbert Space that is convenient for the solution of (54). In [1], the authors consider

$$W^2[-1, 1] = \{u(x) | u^{(1)}(x) \text{ is absolutely continuous, and } u^{(2)}(x) \in L_2[-1, 1]\},$$

provided with the interior product

$$\langle u, v \rangle = \sum_{i=0}^1 u^{(i)}(-1)v^{(i)}(-1) + \int_{-1}^1 u^{(2)}(x)v^{(2)}(x)dx.$$

The Reproducing Kernel is not given explicitly by the authors, but the derivation can be seen in Appendix B. The expression for the kernel is

$$R(x, y) = \begin{cases} \frac{7}{3} - \frac{1}{6}y^3 + \frac{1}{2}y^2x + \frac{3}{2}y + 2xy + \frac{3}{2}x, & y \leq x \\ \frac{7}{3} - \frac{1}{6}x^3 + \frac{1}{2}x^2y + \frac{3}{2}x + 2xy + \frac{3}{2}y, & y > x. \end{cases} \quad (58)$$

Equation (54) is now translated into the space $W^2[-1, 1]$ through the change of variables.

$$t \rightarrow \frac{\gamma(1-\tau)}{1+\tau}, \quad s \rightarrow \frac{\gamma(1-z)}{1+z}, \quad \gamma \in \mathbb{R}^+,$$

$$G(\tau) = Y(\tau) + 2\gamma \int_{-1}^1 \frac{K(\tau, z)}{(z+1)^2} Y(z) dz, \quad (59)$$

where

$$K(\tau, z) = k \left(\frac{\gamma(1-\tau)}{1+\tau} - \frac{\gamma(1-z)}{1+z} \right), \quad Y(\tau) = y \left(\frac{\gamma(1-\tau)}{1+\tau} \right), \quad G(\tau) = g \left(\frac{\gamma(1-\tau)}{1+\tau} \right).$$

There are singularities in (59) along $\tau = z$ as $\tau \rightarrow -1$. The singularities can be subtracted by rewriting the equation in the following manner

$$G(\tau) = \left[(\tau+1)^2 + 2\gamma \int_{-1}^1 K(\tau, z) dz \right] X(\tau) + 2\gamma \int_{-1}^1 K(\tau, z) (X(z) - X(\tau)) dz, \quad (60)$$

with the definition

$$X(z) \equiv \frac{Y(z)}{(z+1)^2}.$$

By noticing that the right-hand side of (60) can be seen as a linear operator acting on $X(\tau)$, we can rewrite the transformed Wiener-Hopf equation as an operator equation.

$$(\mathbb{L}u)(\tau) = G(\tau), \quad -1 < \tau \leq 1, \quad (61)$$

whose solution $u(\tau)$ will be the $X(\tau)$ that satisfies (60), and where

$$(\mathbb{L}u)(\tau) = \left[(\tau+1)^2 + 2\gamma \int_{-1}^1 K(\tau, z) dz \right] u(\tau) + 2\gamma \int_{-1}^1 K(\tau, z)(u(z) - u(\tau)) dz. \quad (62)$$

In [1] it is shown that if $\{x_i\}_{i=1}^\infty$ is any dense sequence in $[-1, 1]$, then, the family of functions $\{\psi_i(\tau)\}_{i=1}^\infty$, given by $\psi_i(x) = \mathbb{L}^* R_{x_i}(x)$ ($\mathbb{L}^*$ denotes the Hermitian conjugate of $\mathbb{L}$) is a complete system in the Hilbert space. It is possible then to transform it to an orthonormalized family of functions $\{\bar{\psi}_i(\tau)\}_{i=1}^\infty$ by the Gram-Schmidt orthogonalization process. The change of basis coefficients β_{ik} from the base $\bar{\psi}$ to the base ψ can be obtained using the inner product.

$$\bar{\psi}_i(x) = \sum_{k=1}^i \beta_{ik} \psi_k(x), \quad (63)$$

$$\psi_i(x) = \sum_{k=1}^i B_{ik} \bar{\psi}_k(x), \quad (64)$$

and using orthogonality of the base $\{\bar{\psi}_i(\tau)\}_{i=1}^\infty$, by Parseval's identity:

$$\langle \psi_i(x), \psi_i(x) \rangle = \sum_{k=1}^{i-1} B_{ik}^2 + B_{ii}^2, \quad (65)$$

$$B_{ii} = \sqrt{\langle \psi_i(x), \psi_i(x) \rangle - \sum_{k=1}^{i-1} B_{ik}^2}, \quad (66)$$

$$\beta_{ii} = \frac{1}{\sqrt{\langle \psi_i(x), \psi_i(x) \rangle - \sum_{k=1}^{i-1} B_{ik}^2}}, \quad (67)$$

and since this system of linear equations is triangular:

$$\beta_{ij} = \beta_{ii} \left(- \sum_{k=j}^{i-1} B_{ik} \beta_{kj} \right). \quad (68)$$

Finally, the solution of (61) is given by

$$X(\tau) = \sum_{i=1}^{\infty} \sum_{k=1}^i \beta_{ik} G(x_k) \bar{\psi}_i(\tau). \quad (69)$$

To recover the solution of (54), we first obtain $Y(\tau) = (\tau + 1)^2 X(\tau)$, and then revert the variable change $\tau \rightarrow \frac{\gamma-t}{\gamma+t}$. In practice, one takes the truncated series up to some number m

$$X(\tau) \approx \sum_{i=1}^m \sum_{k=1}^i \beta_{ik} G(x_k) \bar{\psi}_i(\tau). \quad (70)$$

and revert the change of variable in the same way. A higher choice of m will have greater precision. As for the choice of the parameter γ , the authors recommend that if the interval of estimation is $[0, T]$, then γ should be chosen as 70% of T . By applying this method with both $k(t)$ and $g(t)$ equal to $\Phi(t)$, the solution of the Wiener-Hopf equation will yield an estimate of $f(t) \equiv \alpha h(t)$.

3.5 Direct estimation of α

One of the most widely used measures of variability in spike trains is the Fano factor. Its standard estimator is the ratio of sample variance to sample mean of spike counts observed in a time window W [23].

$$F_W = \frac{\sigma_W^2}{\mu_W}. \quad (71)$$

The quality of the estimator changes with the length of the window W and becomes exact in the limit

$$F = \lim_{W \rightarrow \infty} F_W. \quad (72)$$

In this section, we show that the reproduction number α of the Hawkes process can be expressed in terms of the Fano factor of the spike train $\mathbb{S}$.

Proposition 4. (Direct estimation of α). Let $\mathbb{S}$ be a spike train generated by a Hawkes process with reproduction number α . Then, an estimator for the reproduction number $\hat{\alpha}$ is given by

$$\hat{\alpha} = 1 - \sqrt{\frac{1}{F}}, \quad (73)$$

where F is the Fano factor of the spike train.

Proof. Let $\mathbb{S}$ be a spike train generated by a Hawkes process with parameter α . We want to compute

$$F_W = \frac{\sigma_W^2}{\mu_W}.$$

The average number of spikes in a length of window W is $\mu_W = \bar{\lambda}W$. Now, we look for the variability of the spike count:

$$\sigma_W^2 = \mathbb{E} \left(\int_0^W \frac{dN(t)}{dt} \int_0^W \frac{dN(t')}{dt'} \right) - \mu_W^2, \quad (74)$$

$$\sigma_W^2 = \int_0^W \int_0^W \phi(t - t') dt dt'. \quad (75)$$

By doing the change of variable $\tau \rightarrow t - t'$, the integral becomes

$$\int_0^W \int_0^W \phi(t - t') dt dt' = \int_0^W \int_{-t}^{W-t} \phi(\tau) d\tau dt. \quad (76)$$

We now apply Fubini's theorem to change the order of integration

$$\int_0^W \int_{-t}^{W-t} \phi(\tau) d\tau dt = \int_{-W}^0 \int_{-\tau}^W \phi(\tau) dt d\tau + \int_0^W \int_0^{W-\tau} \phi(\tau) dt d\tau. \quad (77)$$

Carrying out the integrals over t , we obtain

$$\int_{-W}^0 \phi(\tau) (W + \tau) d\tau + \int_0^W \phi(\tau) (W - \tau) d\tau = \int_{-W}^W \phi(\tau) (W - |\tau|) d\tau, \quad (78)$$

so the variability of the spike count is given by

$$\sigma_W^2 = \int_{-W}^W \phi(\tau) (W - |\tau|) d\tau. \quad (79)$$

Notice that for a given W , we are cutting off the tail of the covariance density for $t > W$. Since it should happen that $\phi(t) \in L_1(\mathbb{R})$, there should be some real number $R > 0$ such that $\phi(t) \rightarrow 0$ for all $t > R$. Notice now, that if we take $W \gg R$ and so that $\frac{W-|t|}{W} \approx 1$, then we can make the following approximation

$$\int_{-\infty}^{\infty} \phi(\tau) d\tau \approx \int_{-\infty}^{\infty} \phi(\tau) \frac{(W - |\tau|)}{W} d\tau, \quad (80)$$

$$\begin{aligned} \int_{-\infty}^{\infty} \phi(\tau) \frac{(W - |\tau|)}{W} d\tau &= \int_{-\infty}^{-W} \phi(\tau) \frac{(W - |\tau|)}{W} d\tau + \int_{-W}^W \phi(\tau) \frac{(W - |\tau|)}{W} d\tau + \\ &+ \int_W^{\infty} \phi(\tau) \frac{(W - |\tau|)}{W} d\tau. \end{aligned} \quad (81)$$

Notice that the first and third equations on the right hand side of equation (81) can be neglected since $\phi(t) \rightarrow 0$ for all $t > R$ and $W \gg R$. From this approximation we get

$$\int_{-\infty}^{\infty} \phi(\tau) d\tau \approx \int_{-\infty}^{\infty} \phi(\tau) \frac{(W - |\tau|)}{W} d\tau \approx \int_{-W}^W \phi(\tau) \frac{(W - |\tau|)}{W} d\tau, \quad (82)$$

so from (79) this results in

$$\int_{-\infty}^{\infty} \phi(\tau) d\tau \approx \int_{-W}^W \phi(\tau) \frac{(W - |\tau|)}{W} d\tau = \frac{1}{W} \sigma_W^2. \quad (83)$$

To finalize this proof, consider the following relation proved in [3]:

$$\hat{\phi}(\omega) = \frac{\bar{\lambda}}{|1 - \hat{f}(\omega)|^2}, \quad (84)$$

where $\hat{\phi}(\omega)$ is the Fourier transform of the covariance density, and $\hat{f}(\omega)$ is the Fourier transform of the kernel function. Evaluating (84) in $\omega = 0$ we obtain

$$\int_{-\infty}^{\infty} \phi(t) dt = \hat{\phi}(0) = \frac{\bar{\lambda}}{(1 - \alpha)^2}. \quad (85)$$

Solving for α we get

$$\alpha = 1 - \sqrt{\frac{\bar{\lambda}}{\int_{-\infty}^{\infty} \phi(t) dt}}, \quad (86)$$

and substituting $\int_{-\infty}^{\infty} \phi(t)dt$ from (83), this becomes

$$\alpha \approx 1 - \sqrt{\frac{\bar{\lambda}W}{\sigma_W^2}} = 1 - \sqrt{\frac{\mu_W}{\sigma_W^2}}. \quad (87)$$

The term in the square root can be written in terms of the Fano factor of the spike train to get an estimator of $\hat{\alpha}$:

$$\alpha \approx 1 - \sqrt{\frac{1}{F_W}}. \quad (88)$$

Taking the limit $W \rightarrow \infty$, we finally obtain

$$\hat{\alpha} = 1 - \sqrt{\frac{1}{F}},$$

which completes the proof. $\square$

The previous proposition indicates how to find an estimate of α from F_W for a long enough W . All that is left is to compute F_W from a given spike train $\mathbb{S}$. The expressions for the estimation of F_W can be consulted in [24] and are given by:

$$\hat{\mu}_W = \frac{1}{N} \sum_{k=1}^N N_k(W), \quad (89)$$

$$\hat{\sigma}_W^2 = \frac{1}{N-1} \sum_{k=1}^N \left(N_k(W) - \frac{1}{N} \sum_{k=1}^N N_k(W) \right)^2, \quad (90)$$

$$\hat{F}_W = \frac{\hat{\sigma}_W^2}{\hat{\mu}_W}, \quad (91)$$

where we have divided the spike train in windows of length W and $N_k(W)$ is the number of spikes in the k -th window. This process can be seen schematically in the figure below.



Figure 6: Estimating the Fano factor from the spike count mean and variance of a given spike train.

This finalizes the presentation of the estimation methods. In the following sections we show the results of doing the estimation to simulated data sets.

4 Application to synthetic data

In this section, the performance of the proposed density estimation methods is studied for some example distributions and a suggestion is made regarding which spline order should be chosen. The method is used then to estimate the covariance density of synthetic Hawkes spike trains. The process for generating the data is explained and a goodness of fit test is applied to the generated data to confirm that they indeed follow a Hawkes process.

4.1 Performance of the density estimation methods

In this work, four different kinds of random variable are used for the analysis of performance: an exponential $\text{Exp}(1)$ variable, a $\text{Gamma}(2, 3)$ variable, and two other variables with arbitrary shape, whose probability density functions are given by

$$\begin{aligned} f(x) &= e^{-x} \mathbb{I}_{[0, \infty)}(x), \\ f(x) &= \frac{x}{9} e^{-\frac{x}{3}} \mathbb{I}_{[0, \infty)}(x), \\ f(x) &= \begin{cases} \frac{6}{5}x & x < \frac{1}{2} \\ \frac{9}{10} + \frac{4}{5}(x - \frac{1}{2})^3 & \frac{1}{2} \leq x \leq 1, \end{cases} \\ f(x) &= \begin{cases} \frac{1}{2}x & 0 \leq x < 1 \\ 1 - \frac{1}{2}e^{-(x-1)} & 1 \leq x, \end{cases} \end{aligned}$$

respectively. The reason these distributions were chosen is because they allow us to present key features of our estimation method which are explained below. Some example estimation histograms made with the method proposed in this work are shown below for splines of order $r = 0$ (bar), $r = 1$ (line), $r = 2$ (quadratic), and $r = 3$ (cubic).

Density estimation for some example cases

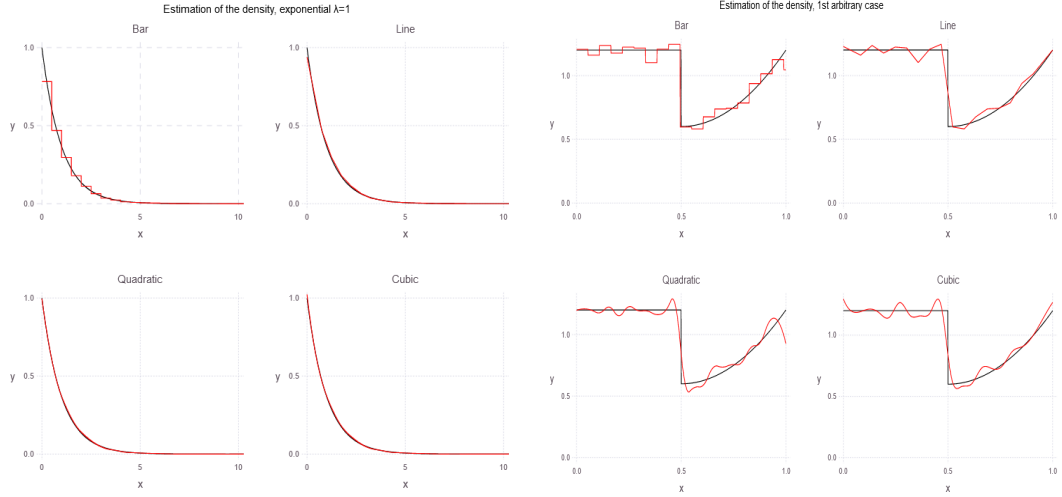


Figure 7: Estimation of the density for the four example cases. The black line is the graph of theoretical pdf, and the red line is the estimation. Sample size is $N = 10^4$ and the bandwidth is not necessarily the optimal choice Δ^* .

For making a comparison of the performance between the methods proposed, each one of them is used for estimating the density of simulated data sets of varying size N from each of the above distributions. Since the underlying distribution is known, a computation of the MISE between the estimated density function and the real density is done directly for each value of N and for a wide range of values of the bandwidth parameter Δ . The MISE at the optimal Δ^* is stored for each value of N . To account for the variability that is intrinsic to random numbers, M samples are generated for each value of N and the MISE is averaged over those M trials. The standard deviation is also computed to measure the variability of the MISE. The results from this analysis for splines of order $r = 0, 1, 2, 3$ and the Parzen window method with a square and gaussian window functions are summarized in the following figure.

Scaling of the MISE with the sample size N

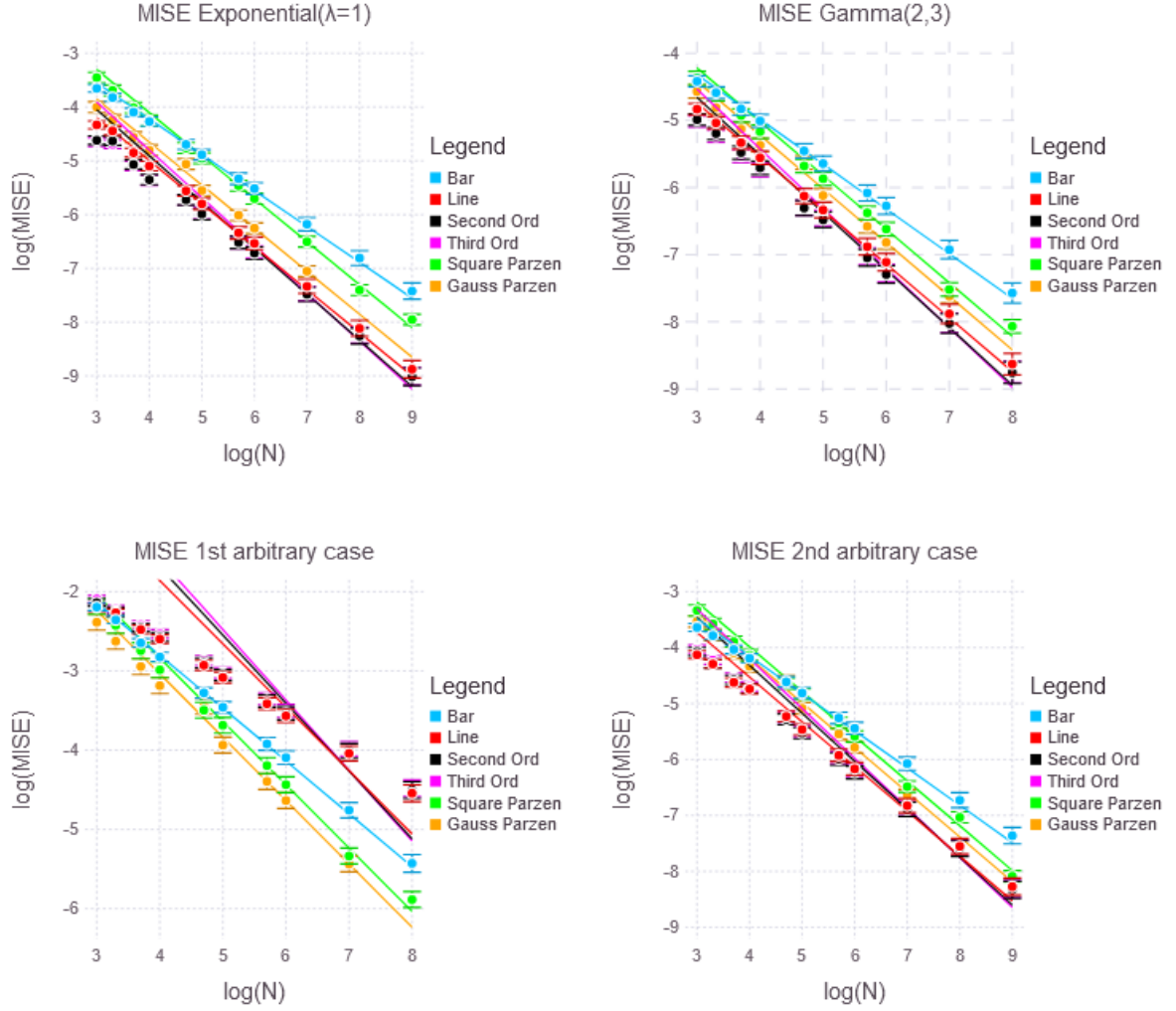


Figure 8: Behavior of the MISE for each of the estimation methods. The solid lines show the asymptotic behavior of the MISE as power laws $N^{-\gamma}$ with $\gamma = \frac{2}{3}, \frac{4}{5}, \frac{6}{7}, \frac{8}{9}$ for splines of order $r = 0, 1, 2, 3$ respectively and $\gamma = \frac{4}{5}$ for both of the Parzen window methods. Notice the inconsistency with the arbitrary cases. Each point was computed with $M = 30$ samples.

Notice that for the gamma and exponential cases, the MISE follows the predicted scaling law, but for the arbitrary cases there is a deviation. These deviations occur because the distributions used do not satisfy the differentiability conditions that we require. In the case of the first arbitrary case, there is a discontinuity, and thus, no derivative of any order. Any spline histogram of order greater than zero will not follow the scaling law. For the case of the second arbitrary case, even when there is no derivative because of the sharp edge, it could be considered close enough to a smooth function and hence the spline histograms of zeroth and first order follow our scaling laws. For the case of greater order, they start to deviate from our prediction. If we want to reliably use our method, we can look at the original bar histogram and look for possible discontinuities or sharp edges. If there were some, then we would have to stick to constant or linear splines.

4.2 Generation of Hawkes Spike Trains

In order to generate a Hawkes spike train, it is necessary to first fix the value of the parameters ρ , α and the kernel function $h(t)$ making sure that they satisfy the required conditions. The algorithm for generating a spike train $\mathbb{S}$ with N elements should give the time-stamps of the generated events and update the value of the firing rate.

The process for generating the spikes can be understood by looking at the limit definition of the firing rate

$$\lambda(t) = \lim_{\Delta t \downarrow 0} \frac{\mathbb{E}\{N(t + \Delta t) - N(t) | H_t\}}{\Delta t},$$

so for small Δt , the average number of events in an interval Δt is really close to $\lambda(t)\Delta t$. This principle is the foundation of the following algorithm to generate the synthetic data for this work.

Algorithm (Generating a spike train of N elements).

1. *Set up initial values for the number of spikes $M = 0$ and time $T = 0$. Choose a fixed value for a small increment dt . Create a collection variable $\mathbb{S}$ to store the time-stamps of the events.*
2. *Evaluate $\lambda(t)$ at a given value of T and M and compute $\lambda(t) \cdot dt$.*
3. *Generate a continuous uniform $U[0, 1]$ random number u .*
4. *Make the comparison $u \leq \lambda(t)dt$. If it is true, increase $M = M + 1$ and store time T in $\mathbb{S}$; if it is false, leave $M = M$ and do not store the time.*
5. *Update the value of the time by $T = T + dt$.*
6. *Repeat steps 2 through 5 until the spike train has the desired amount of elements $M = N$. Collection $\mathbb{S}$ will be a Hawkes spike train.*

It can be confirmed that the spike trains generated by this algorithm correctly represent a Hawkes process. This is done by using *time scaling theory* to analyze the times between consecutive spikes also called *inter-arrival times*. The time scaling relies on the following theorem.

Theorem 2. (Adapted from [9], p.258, Theorem 7.4.I).

Let $N(t)$ be the counting process associated to a Hawkes process with event history H_t , strictly positive firing rate $\lambda(t)$ and with $\Lambda(t) = \int_0^t \lambda(s)ds$ that is not almost surely bounded. If $\tau \equiv \Lambda(t)$, then under the random time change $t \rightarrow \tau$, the transformed process defined as

$$\tilde{N}(t) = N(\Lambda^{-1}(t)) = N(\tau), \quad (92)$$

is a homogeneous Poisson process with unit rate.

This theorem establishes that if we have a spike train $\mathbb{S} = \{t_1, t_2, \dots, t_n\}$ that comes from a Hawkes process, and scale the inter-arrival times to $\tilde{\mathbb{S}} = \{\tau_1, \tau_2, \dots, \tau_{n-1}\}$ where

$$\tau_i = \int_{t_i}^{t_{i+1}} \lambda(t)dt, \quad (93)$$

then $\tilde{\mathbb{S}} = \{\tau_1, \tau_2, \dots, \tau_{n-1}\}$ will have the same distribution as the inter-arrival times of a homogeneous Poisson process with intensity one.

It is a known result that the inter-arrival times s_i for such a Poisson process are exponentially distributed with parameter one, i.e. $s_i \sim \text{Exp}(1)$. The goodness of fit test involves then comparing the distribution of the inter-arrival times $\tilde{\mathbb{S}}$ against an exponential distribution with parameter one, with probability density function $f(s) = e^{-s}\mathbb{I}_{[0,\infty)}(s)$.

One procedure for comparing two probability distributions is to plot their respective quantiles against each other. Such a plot is known as a *quantile-quantile plot* (Q-Q plot). If the compared distributions are similar, then the points in the plot will lie close to the line $y = x$. In order to draw this plot, a collection of $n - 1$ exponentially distributed points is generated and arranged in ascending order. The points from empirical data in $\tilde{\mathbb{S}}$ are also arranged in ascending order and plotted against the exponentially distributed points. This is shown in the figure below for empirical data generated using our algorithm for exponential and square kernel functions.

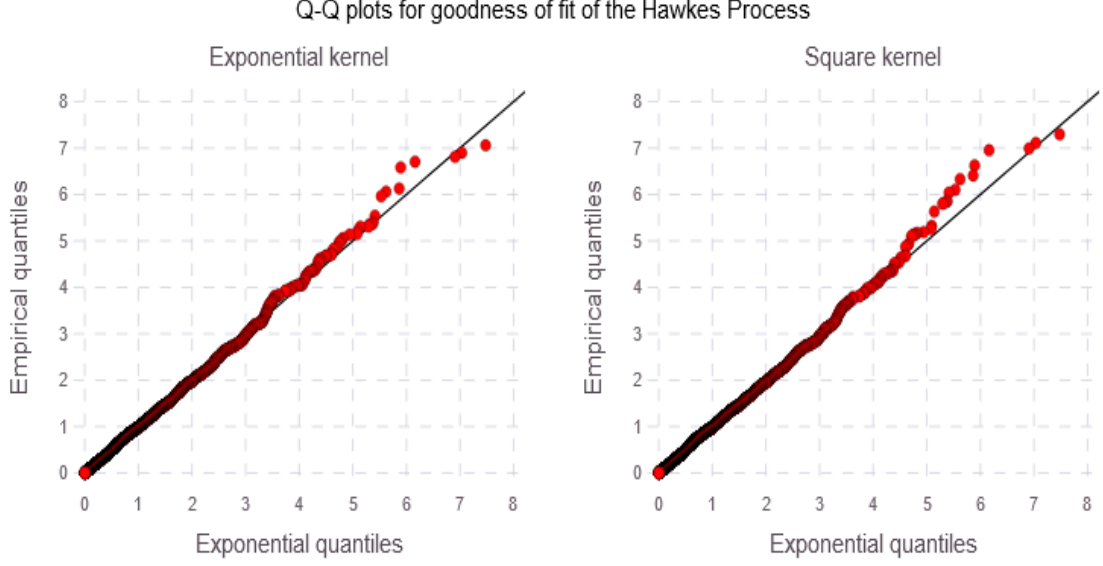


Figure 9: Goodness of fit test for Hawkes processes with parameters $\rho = 1$, $\alpha = 0.7$, and exponential $h(t) = \frac{1}{\tau}e^{-\frac{t}{\tau}}\mathbb{I}_{[0,\infty)}(x)$ and square $h(t) = \frac{1}{\tau}\mathbb{I}_{[0,\tau]}$ kernels with time scale $\tau = 10$. $N = 3 \cdot 10^3$ samples. It can be seen that in both cases the points lie approximately on the line $y = x$, which confirms that the spikes come indeed from a Hawkes process.

The Q-Q plots show that the presented algorithm is suitable for simulating Hawkes spike trains.

4.3 Empirical laws for choosing the optimal Δ^*

Since we have an algorithm to generate Hawkes spike trains, we proceed to the determination of the constants C_r to find the empirical laws (39) for splines up to third order. For this, we generated a collection of spike trains from a Hawkes process with reproduction number $\alpha \in \{0.1, 0.2, \dots, 0.9\}$ and exponential kernels of time scale $\tau \in \{1, 5, 10, 15, 20\}$. The sample consisted of 5 spike trains for each combination of (τ, α) with $N = 10^4$ and $N = 10^5$ for each case. The explicit form of the exponential kernel is

$$f(t) = \frac{\alpha}{\tau}e^{-\frac{t}{\tau}}\mathbb{I}_{[0,\infty)}(t). \quad (94)$$

The covariance can be derived explicitly as shown in [17]

$$\phi(t) = \frac{\alpha\rho(2-\alpha)}{2\tau(1-\alpha)^2}e^{-(1-\alpha)\frac{|t|}{\tau}}\mathbb{I}_{\mathbb{R}}(t), \quad (95)$$

and by using expression (6), it can be written as

$$\phi(t) = \frac{\alpha(2-\alpha)}{2\tau(1-\alpha)}\bar{\lambda}e^{-(1-\alpha)\frac{|t|}{\tau}}\mathbb{I}_{\mathbb{R}}(t). \quad (96)$$

From this, the theoretical width can be computed as

$$\sigma = \frac{\sqrt{2}\tau}{(1-\alpha)}. \quad (97)$$

For each spike train, the covariance density was computed with our method for several values of the bin size Δ . The Mean Integrated Squared Error was then computed between our estimation for each value of Δ and the theoretical function (96). The optimal Δ^* was chosen as the value of Δ that minimized the MISE. Since from (39) we had that

$$\Delta^* = C_r \sigma N^{-\frac{1}{2r+3}}.$$

We can take a logarithm on both sides and write

$$\log\left(\frac{\Delta^*}{N^{-\frac{1}{2r+3}}}\right) = \log(\sigma) + \log(C_r), \quad (98)$$

this means that if we define

$$F(\sigma) \equiv \log\left(\frac{\Delta^*}{N^{-\frac{1}{2r+3}}}\right), \quad (99)$$

then $F(\sigma)$ should change linearly in $\log(\sigma)$ with slope one. By making a plot of $F(\sigma)$ against $\log(\sigma)$, and doing a linear regression with an imposed slope of one, we can get the intersect at the origin $B = \log(C_r)$. We can get the constants C_r by taking the exponential of the estimated intersects, $C_r = e^B$, and write the empirical laws for the selection of the bin-size. An example of this procedure can be seen in figure 10.

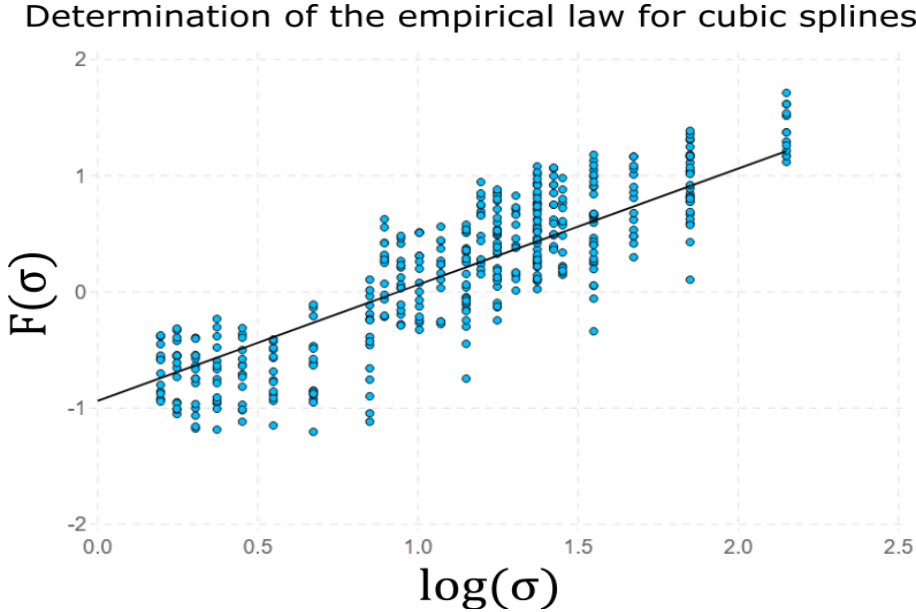


Figure 10: Determination of the empirical laws using a data set corresponding of spike trains with $N \in \{10^4, 10^5\}$ with reproduction number $\alpha \in \{0.1, 0.2, \dots, 0.9\}$ and exponential kernels of time scale $\tau \in \{1, 5, 10, 15, 20\}$ for splines of order $r = 3$.

The empirical laws for splines of order $r = 0, 1, 2, 3$ are

$$\Delta^* = \begin{cases} 8.05\sigma N^{-\frac{1}{3}} & r = 0 \\ 0.04\sigma N^{-\frac{1}{5}} & r = 1 \\ 0.12\sigma N^{-\frac{1}{7}} & r = 2 \\ 0.19\sigma N^{-\frac{1}{9}} & r = 3, \end{cases} \quad (100)$$

which allow to choose the bandwidth by specifying the order of the spline r , the number of spikes N and the width σ . As we indicated in previous sections, the value of the estimator $\hat{\sigma}$ should be used instead of the theoretical value.

The computation of the estimator $\hat{\sigma}$ can be implemented as an algorithm with complexity $O(N)$, and for a sample size of $N = 10^4$ the computation time is of the order of one second. This proves to be faster than the cross-validation algorithm used in [6] which has complexity $O(N \log(N))$ and the computation time for a sample size $N = 10^4$ is of the order of tens of seconds. Moreover, the computation times of a spline histogram and a Parzen window estimate with a square kernel are comparable, but with a more sophisticated kernel, the computation time increases rapidly as explained in [27].

From this observation and that of Section 4.1, we can conclude that our method provides estimates of the covariance density with faster decrease of the MISE for $r \geq 1$, and smaller absolute MISE for $r \geq 2$. The estimates as well are computed faster as shown by the comparison of computation time.

4.4 Performance of the density estimation methods

We analyzed the performance of both estimation methods. We will say that one method performs better than the other if the estimation MISE is smaller. At first sight, it is visible that our method provides an estimation that looks smoother and closer to the original function. One example of this can be seen in figure 11, where we applied both methods to a spike train with $N = 10^5$ from a Hawkes process with exponential kernel $f(t) = 0.07e^{-0.1t}$. For our method, we used second order splines ($r = 2$).

To precisely compare the performance of our kernel estimation method with that of [6], we used both approaches to estimate the kernel of a data set of 150 spike trains generated according to Hawkes processes with $\rho = 1$, and exponential kernel with reproduction number and time scale (α, τ) selected at random with equal probabilities from the possible values $\alpha \in (0, 1)$ and $\tau \in (1, 20)$. For each spike train, we took $N = \{10^3, 10^4, 5 \times 10^4, 10^5, 5 \times 10^5, 10^6\}$ events. We computed the MISE between the estimate $\hat{f}(t)$ and the real kernel function $f(t)$ for every sample for each value of N . The average MISE was taken in each case and the results are summarized in figure 12.

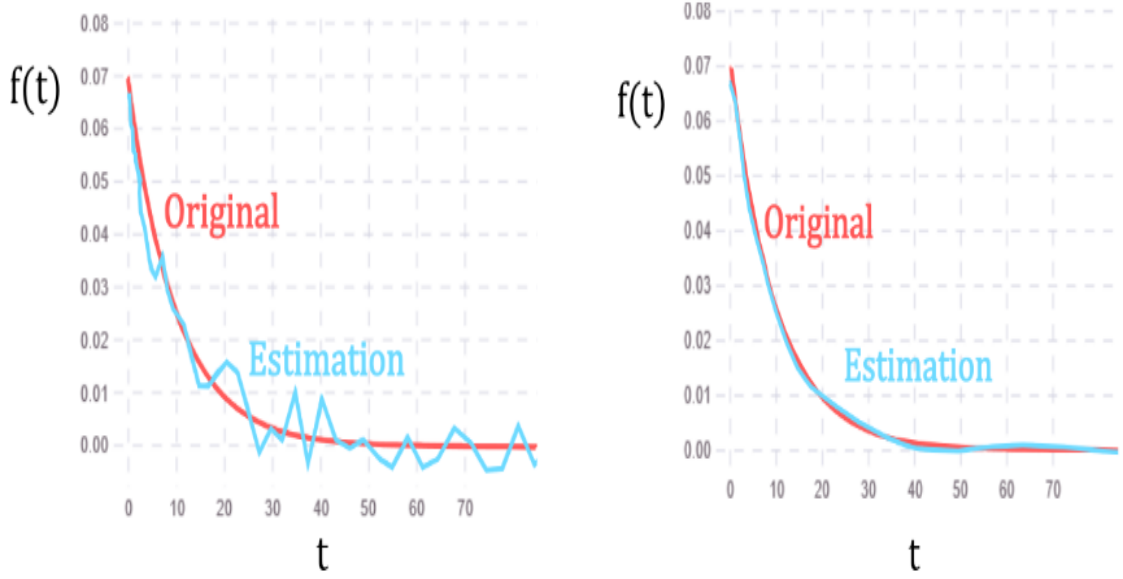


Figure 11: Estimation of the kernel with the existing method (left) and with our method (right).

Performance comparison for the estimation methods

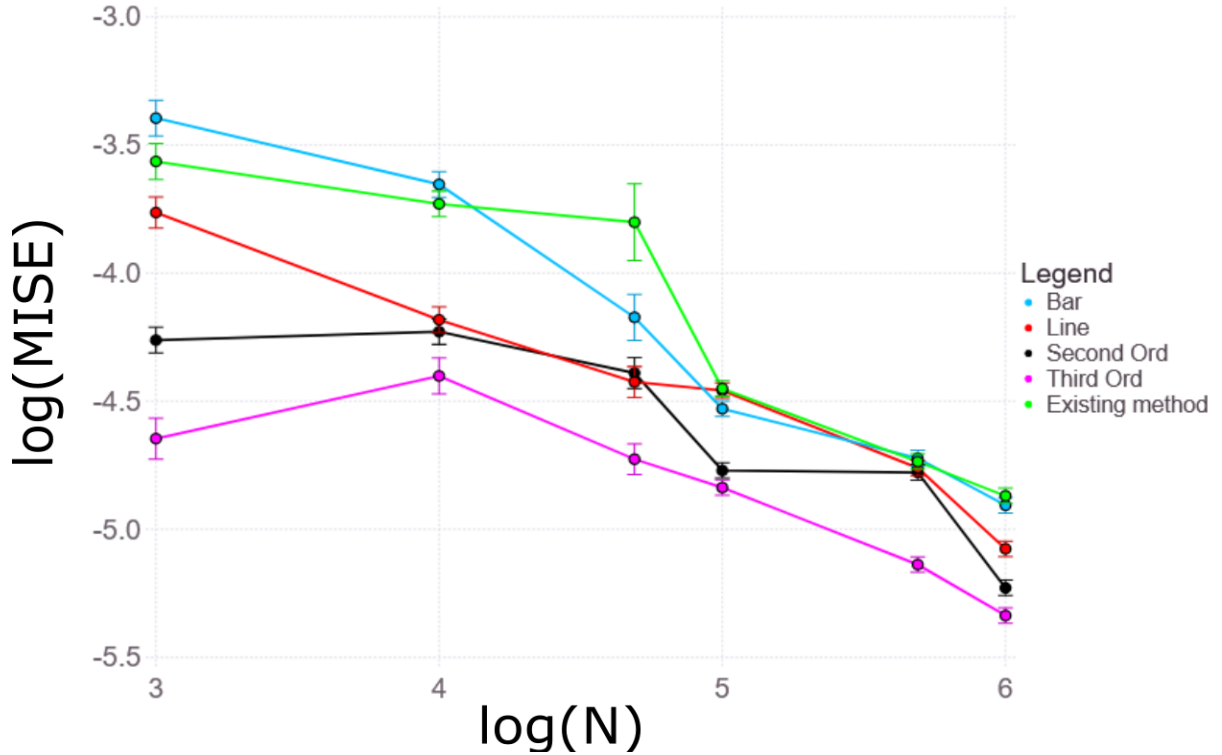


Figure 12: Comparison of the average estimation MISE for the existing method and our method with splines up to third order.

We can see that for large N , specially $N \geq 10^5$, our method gives some improvement compared to the existing one. This can be seen as a smaller MISE between $\hat{f}(t)$ and $f(t)$. On the other hand, we can see that for smaller N , specially $N \leq 10^4$, the existing

method gives a better estimation than ours with constant splines ($r = 0$), but for splines of order $r \geq 1$, our method becomes superior. The *MISE* for splines of order $r = 3$ is at least one order of magnitude smaller than in the existing method for $N = 10^3$.

4.5 Estimation of α from the Fano factor

We showcase our estimation of the reproduction number by applying it to a Hawkes process with an exponential kernel and reproduction number of $\alpha = 0.6$, this is shown in figure 13. Notice that expression (73) does not depend on the shape of the kernel directly. Nevertheless, we are limited by the number of spikes available in the spike train, since we need to take a long enough window W to ensure that approximation (82) is valid.

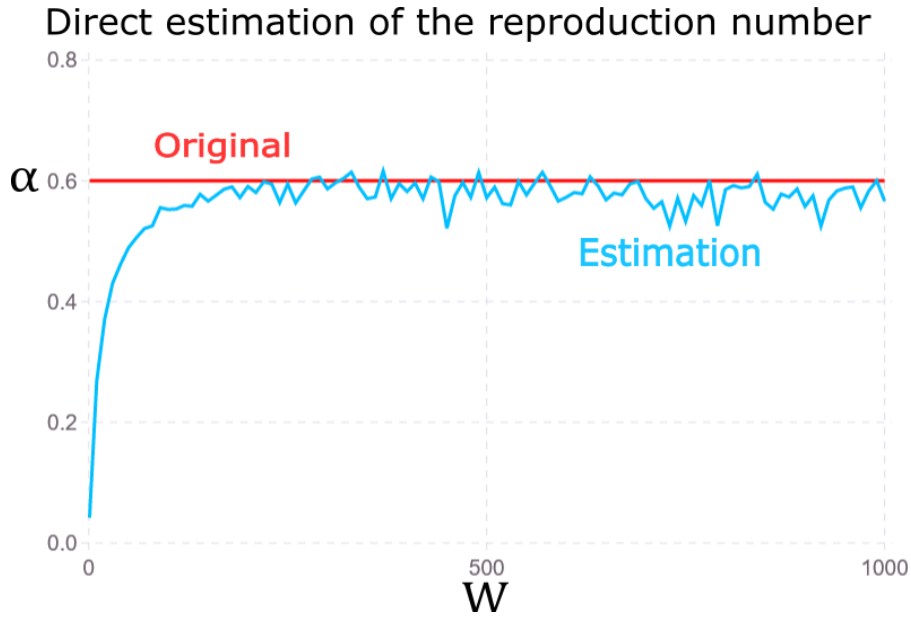


Figure 13: Estimation of the reproduction number α from the Fano factor of the spike train. We can obtain a close estimate by taking a long enough window W .

5 Application to real data

We applied our method to data obtained from neuronal spike trains from the brain of a monkey. These data sets were obtained in a joint collaboration [19] and consisted of several spike trains for each brain region: Higher-order motor (Pre-SMA), primary motor cortex (M1), primary visual cortex (V1) and pre-frontal cortex (PF). Each spike train contained exactly 2000 spikes measured in the scale of milliseconds (ms). The estimated kernels can be seen in the figure below.

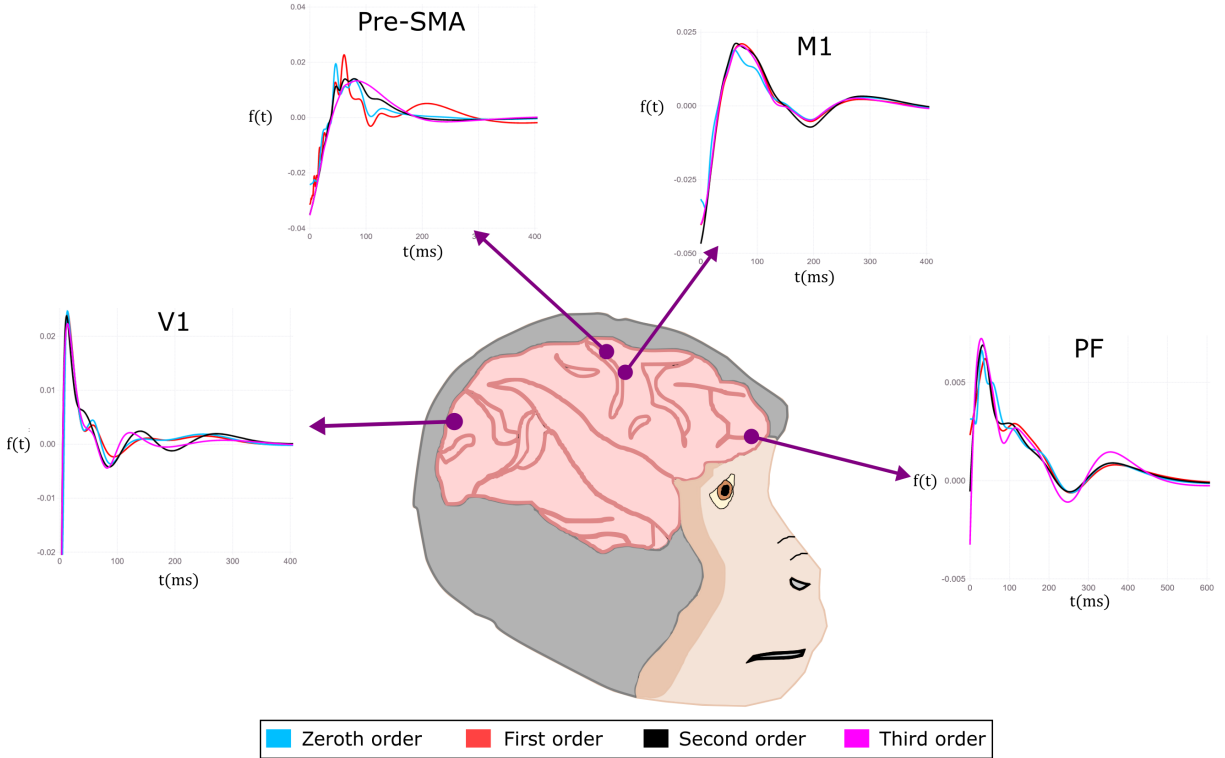


Figure 14: Estimation of the kernel from neuronal spike data of a monkey for the Motor (M1), Higher-order motor (Pre-SMA), Visual (V1) and Pre-frontal (PF) brain regions.

We can see that all kernels take negative values within the first 50 milliseconds. This is consistent with what is observed in biological neurons, which have a refractory period of inactivity before they are able to fire again. It can be seen as well that the use of higher order splines provides a smoother kernel. When analyzing these data, our method provided results that were consistent with each other between all the choices of spline orders $r = 0, 1, 2, 3$.

Now that we found an estimate for the kernel functions from neural data, we can also make an estimation of the reproduction number from the Fano factor of the spike train and from the integration of the kernels as suggested in equation (41). We expect that both approaches yield a similar result. In figure 15 we plotted, the reproduction number found using the Fano factor (α_F) and the ones found by integrating the kernels computed with splines of i -th order (α_i). We can see that, as expected, data points clump near the line with slope one.

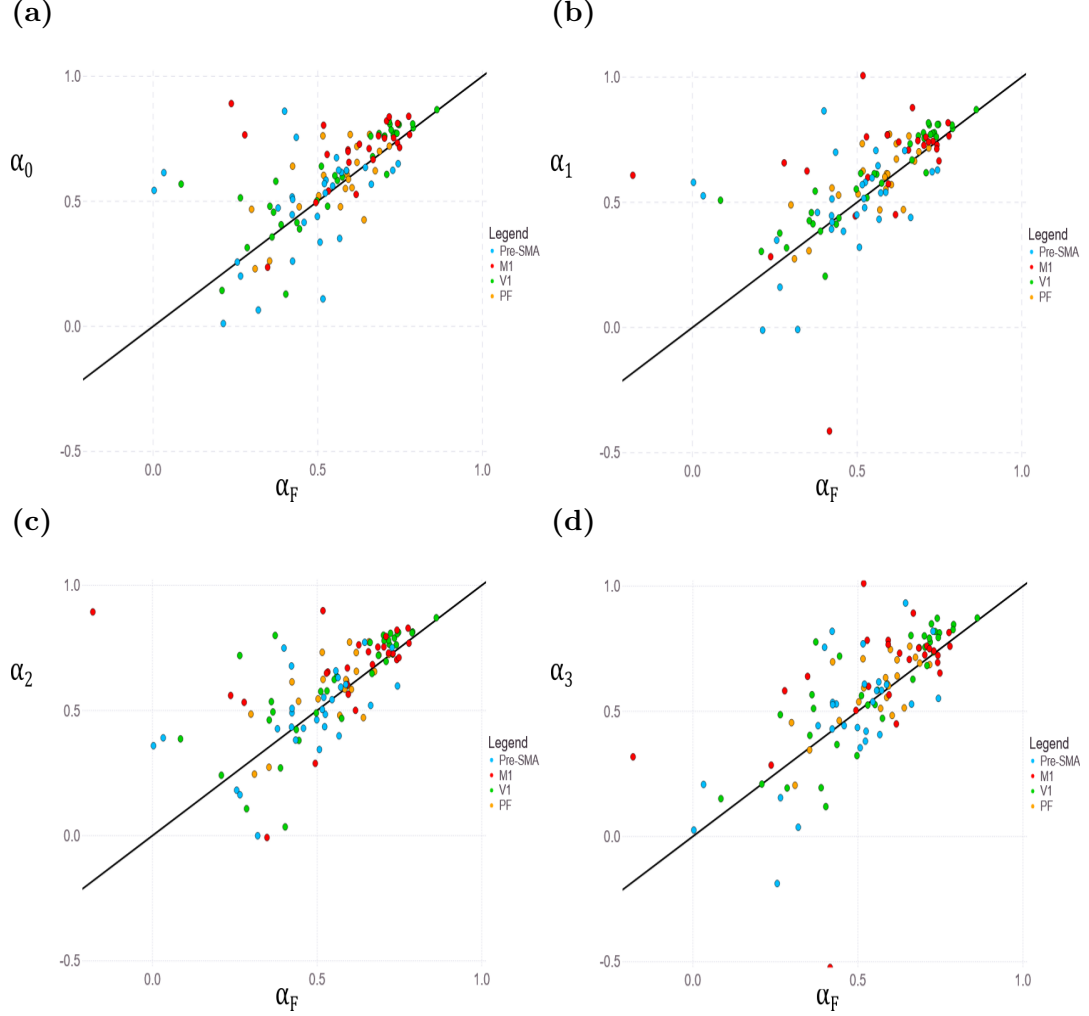


Figure 15: Comparison of the reproduction number estimated with the Fano factor (α_F) and integrating the kernels using splines of i -th order (α_i) for: **(a)** Zeroth order **(b)** First order **(c)** Second order and **(d)** Third order.

6 Discussion

Nowadays, it is very common to work with large volumes of data. As technology improves, it becomes easier to acquire long measurements from event occurrences in diverse phenomena, and many times these data sets are made publicly available. We have developed a method, based on the Hawkes model, to study the self-exciting nature of these phenomena. We have to assume that data is stationary (or approximately stationary), but since it is a non-parametric approach, we do not have to assume any predetermined shape of the kernel function. As we have shown, our method is reliable enough to be used in practice.

6.1 Limitations

As we presented it at the beginning of this work, the Hawkes process is restricted to the stationary case. Because of this, when analyzing spike trains, the quality of the estimation will be affected by whether the data is stationary or not. In the case of synthetically generated data, we made sure that the spike train was stationary by simulating a longer spike train than needed and removing the spikes from some initial time interval. Of course, we are also limited by the number of spikes in both the estimation of the kernel and the direct estimation of the reproduction number, as we previously mentioned.

When analyzing the asymptotic behavior of our density estimators, if the density function fails to be continuous, or have a derivative at some point or interval, then the predicted scaling is not correct. In practice, one does not usually encounter highly pathological cases, and single point discontinuities can be ignored if they are avoidable, or treated by bar histograms if they are jump discontinuities.

6.2 Possible improvements

We developed a method for selecting the optimal bin size Δ^* for the estimation of the covariance density. As we confirmed with both real and synthetic data, this empirical rule yields a reasonable estimation. However, as seen in figure 10, there is still some variability that we are not capturing in our selection rule and that makes the data points distribute around the linear regression curve. A possible improvement for our method would be to determine where this variability comes from, so that we can add a correction term to our empirical law and do more accurate estimates.

In the case of the direct estimation of α , we can see in figure 13 that there are several choices of W that yield a good estimate. In fact, we can see that there is an interval $[W_1, W_2]$ from where we can choose a good W . Currently, we are first plotting $\hat{\alpha}$ for several choices of W , determining $[W_1, W_2]$ ‘by eye’, and then averaging over the chosen interval. Another possible improvement of our method would be to standardize a method for choosing $[W_1, W_2]$ automatically.

6.3 Conclusion

We have developed a method for estimating the self-exciting properties of a Hawkes process from event series data. Our method proved to be superior in both estimation performance and computation speed to existing methods. Our method also allows us to change the accuracy and smoothness of the estimation by changing the order of splines.

This gives our approach adaptability when encountering some pathological cases, as analyzed before, or when computation time is a constraint. Finally, we showed that the reproduction number can be directly estimated from the event series in a simple way, and that this estimation can be used to validate our kernel estimation. Overall, our method is fast and simple to use, which makes it convenient when analyzing event series data.

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8 Appendix A (Inequality for equally spaced nodes)

In the proof of **Proposition 2**, we indicated that we could bound $|w_{r+1}(x)|$ in the following way:

$$\max |w_{r+1}(x)| = \max \left| \prod_{j=0}^r (x - x_j) \right| \leq \Delta^{r+1} \frac{1}{4} r!. \quad (101)$$

Let us prove this statement.

Proof. Without loss of generality, let us suppose that for the index i we have $x \in [x_i, x_{i+1}]$. Then, for some $q \in [0, 1]$ we can write $x = x_i + q\Delta$. Notice that we can split the product in the following way:

$$\prod_{j=0}^r |(x - x_j)| = \prod_{j=0}^{i-1} |(x - x_j)| \cdot |(x - x_i)(x - x_{i+1})| \cdot \prod_{j=i+2}^r |(x - x_j)|.$$

Notice that for $\prod_{j=0}^{i-1} |(x - x_j)| \Rightarrow |x - x_j| < (i - j + 1)\Delta$, for $\prod_{j=i+2}^r |(x - x_j)| \Rightarrow |x - x_j| \leq (j - i)\Delta$, and for the middle terms $|(x - x_i)(x - x_{i+1})| = \Delta^2 |q(1 - q)| \leq \frac{1}{4}\Delta^2$. By multiplying everything, we get:

$$\prod_{j=0}^r |(x - x_j)| \leq \Delta^{r-1} (i+1)!(n-i)!q(1-q) \leq \frac{1}{4}\Delta^{r+1} (i+1)!(n-i)!.$$

The term $(i+1)!(r-i)!$ is maximum if either $i = 0$ or $i = r - 1$. In both cases $(i+1)!(r-i)! = r!$, so we obtain

$$\max \left| \prod_{j=0}^r (x - x_j) \right| \leq \Delta^{r+1} \frac{1}{4} r!,$$

which completes the proof. □

9 Appendix B (Computation of the reproducing kernel)

Here, we present the computation of the Reproducing Kernel for the Hilbert Space. Let us reiterate that we are working in the space

$$W^2[-1, 1] = \{u(x) | u^{(1)}(x) \text{ is absolutely continuous, and } u^{(2)}(x) \in L^2[-1, 1]\},$$

equipped with the interior product

$$\langle u, v \rangle = \sum_{i=0}^1 u^{(i)}(-1)v^{(i)}(-1) + \int_{-1}^1 u^{(2)}(x)v^{(2)}(x)dx.$$

From the definition of the reproducing kernel, we know that if $u(x) \in W^2[-1, 1]$, then

$$\langle u(y), R_x(y) \rangle = u(x), \quad \forall u \in W^2[-1, 1].$$

This translates to

$$\begin{aligned} u(x) &= \langle u(y), R_x(y) \rangle = u(-1)R_x(-1) + u'(-1)R'_x(-1) + \int_{-1}^1 u''(x)R''_x(y)dy = \\ &= u(-1)R_x(-1) + u'(-1)R'_x(-1) + u'(y)R''_x(y)|_{-1}^1 - u(y)R'''_x(y)|_{-1}^1 + \int_{-1}^1 u(y)R^{(4)}_x(y)dy. \end{aligned}$$

Since $u(x) = \langle u(y), R_x(y) \rangle$, we have:

$$\begin{cases} R_x^{(4)}(y) = \delta(y - x) & (102a) \\ R_x(-1) + R'''_x(-1) = 0 & (102b) \\ R'_x(-1) - R''_x(-1) = 0 & (102c) \\ R'''_x(1) = 0, \quad R''_x(1) = 0. & (102d) \end{cases}$$

This differential equation is $R_x^{(4)}(y) = 0$ for $y \neq x$. The characteristic equation by taking $R_x(y) = e^{\lambda_x y}$ is $\lambda_x = 0$ for all $x \neq y$. The solution will be a polynomial in y , the coefficients will include the x dependency.

$$R_x(y) = \begin{cases} \sum_{i=1}^4 c_i y^{i-1}, & y \leq x \\ \sum_{i=1}^4 d_i y^{i-1}, & y > x. \end{cases} \quad (103)$$

In order to find the coefficients c_i and d_i , we impose continuity conditions on $R_x(y)$ and its derivatives, together with conditions (102a),(102b),(102c) and (102d). We get the set of conditions:

$$\begin{cases} R_x^{(m)}(x+0) = R_x^{(m)}(x-0), & (m = 0, 1, 2) & (104a) \\ R'''_x(x+0) - R'''_x(x-0) = 1 & (104b) \\ R_x(-1) + R'''_x(-1) = 0 & (104c) \\ R'_x(-1) - R''_x(-1) = 0 & (104d) \\ R'''_x(1) = 0 & (104e) \\ R''_x(1) = 0. & (104f) \end{cases}$$

To solve this, let us write $R_x(y)$ and its first three derivatives explicitly:

$$R_x(y) = \begin{cases} c_0 + c_1y + c_2y^2 + c_3y^3, & y \leq x \\ d_0 + d_1y + d_2y^2 + d_3y^3, & y > x, \end{cases} \quad (105)$$

$$R'_x(y) = \begin{cases} c_1 + 2c_2y + 3c_3y^2, & y \leq x \\ d_1 + 2d_2y + 3d_3y^2, & y > x, \end{cases} \quad (106)$$

$$R''_x(y) = \begin{cases} 2c_2 + 6c_3y, & y \leq x \\ 2d_2 + 6d_3y, & y > x, \end{cases} \quad (107)$$

$$R'''_x(y) = \begin{cases} 6c_3, & y \leq x \\ 6d_3, & y > x. \end{cases} \quad (108)$$

From conditions (104e) and (104f) we see that $d_3 = 0$ and $d_2 = 0$ respectively. Note that

$$R'''_x(y) = \begin{cases} 6c_3, & y \leq x \\ 0, & y > x, \end{cases} \quad (109)$$

can be combined with condition (104b) to get

$$R'''_x(x+0) - R'''_x(x-0) = 1 \Rightarrow -6c_3 = 1 \Rightarrow c_3 = -\frac{1}{6},$$

so

$$R'''_x(y) = \begin{cases} -1, & y \leq x \\ 0, & y > x. \end{cases} \quad (110)$$

From these coefficients, we can also see that for the second derivative

$$R''_x(y) = \begin{cases} 2c_2 - y, & y \leq x \\ 0, & y > x. \end{cases} \quad (111)$$

We can apply condition (104a) for $m = 2$: $R''_x(x+0) = R''_x(x-0)$ to find that

$$0 = 2c_2 - x \Rightarrow c_2 = \frac{x}{2},$$

which gives us an explicit way to write the second derivative

$$R''_x(y) = \begin{cases} x - y, & y \leq x \\ 0, & y > x. \end{cases} \quad (112)$$

Applying condition (104d) to the first derivative, we obtain

$$R'_x(-1) = R''_x(-1) \Rightarrow c_1 - x - \frac{1}{2} = x + 1 \Rightarrow c_1 = 2x + \frac{3}{2}.$$

The first derivative becomes

$$R'_x(y) = \begin{cases} \frac{3}{2} + 2x + xy - \frac{1}{2}y^2, & y \leq x \\ \frac{3}{2} + 2x + \frac{1}{2}x^2, & y > x. \end{cases} \quad (113)$$

In addition, we apply condition (104a) for $m = 1$ and we find that

$$d_1 = \frac{3}{2} + 2x + \frac{1}{2}x^2.$$

So far, for $R_x(y)$, we have that

$$R_x(y) = \begin{cases} c_0 + 2xy + \frac{3}{2}y + \frac{x}{2}y^2 - \frac{1}{6}y^3, & y \leq x \\ d_0 + \frac{3}{2}y + 2xy + \frac{1}{2}x^2y, & y > x. \end{cases} \quad (114)$$

We apply condition (104c) and find that

$$c_0 = \frac{3}{2}x + \frac{7}{3},$$

and finally, using condition (104a) for $m = 0$:

$$d_0 = -\frac{1}{6}x^3 + \frac{3}{2}x + \frac{7}{3},$$

which was the last coefficient to be determined. The form of the reproducing kernel for this Hilbert space is then

$$R(x, y) = \begin{cases} \frac{7}{3} - \frac{1}{6}y^3 + \frac{1}{2}y^2x + \frac{3}{2}y + 2xy + \frac{3}{2}x, & y \leq x \\ \frac{7}{3} - \frac{1}{6}x^3 + \frac{1}{2}x^2y + \frac{3}{2}x + 2xy + \frac{3}{2}y, & y > x, \end{cases} \quad (115)$$

which is what we indicated in equation (58).